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ORIGINAL ARTICLE

Applying Data Science Practices to Identify Characteristics of Postsecondary Autism Support Programs From Their Websites

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Abstract

Purpose An increasing number of autistic students in the United States are seeking post-secondary education. In response, some post-secondary institutions have established Autism Support Programs (ASP) to address the comprehensive needs of this population. There is little up-to-date, comprehensive information about which institutions host these programs, what types of services they offer, and what is required to access them.

Methods Expanding on previous research, we introduce a new method, which utilizes established data science techniques, to identify ASPs at post-secondary institutions in the U.S. Our technique also allows us to identify the characteristics of the ASPs, including admissions requirements, cost, structure, and supports offered.

Results Results highlight our method is more efficient and more robust than previous methods from the literature. For example, we identify 49 schools hosting ASPs that were not identified in past literature searches. We report on the characteristics of identified ASPs such as application process, most common supports and program cost.

Conclusion The bi-directional change in the number of ASPs shows that this is an evolving field, requiring automated tools to enable regular updates to data. Although it is promising that a relative handful of U.S. schools have established these programs, a large majority of post-secondary institutions have not, and for those that host them, barriers to access exist, including the necessity of an ASD diagnosis, coupled with up-front and ongoing costs.

Keywords Applied data science · Higher education accessibility · Autism spectrum disorder · College autism support programs

Applying Data Science Practices to Identify Characteristics of Postsecondary Autism Support Programs From Their Websites

The number of children diagnosed with autism spectrum disorder (ASD) has been on the rise, with recent estimates suggesting 1 in 36 children have ASD (Maenner et al., 2023). Research estimates that 0.9% of college freshmen in the U.S. are diagnosed with ASD (Stolzenberg et al., 2019), and others who have ASD remain undiagnosed. Despite greater participation in higher education, individuals with ASD still face significant barriers to accessing a postsecondary degree yet, an increasing number of autistic students are seeking some form of postsecondary education (Bakker et al., 2019), leading some institutions to establish offices and programs known as ASPs (Autistic Support Programs) to address autistic students' unique challenges. These programs aim to provide support tailored to the strengths and challenges unique to this population (Barnhill, 2016).

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ASPs are designed to address a variety of concerns encountered by this population in higher education. For example, many students with ASD receive comprehensive special education services in K-12 education, but these services abruptly end when they graduate from high school, often with inadequate planning for the transition to higher education (Cai & Richdale, 2016). Autistic students in K-12 education are not required to have a formal ASD diagnosis to receive services (Individuals with Disabilities Education Act, 2004), or they may have received a diagnosis many years before completing high school. Those who fail to disclose their diagnosis or lack a formal diagnosis may not receive supports at the postsecondary level (Anderson et al., 2017; Cai & Richdale, 2016; Cox et al., 2017). Although students with ASD can succeed academically, they are also likely to struggle with non-academic challenges such as social isolation, stress, anxiety, and depression (Jackson et al., 2018). These conditions may be exacerbated by secondary diagnoses (e.g., ADHD, depression, anxiety, intellectual disabilities), which are more prevalent with ASD (Hodges et al., 2020). Finally, predictors of college success for students with ASD differ from those for other students (Bakker et al., 2023). Thus, the kinds of typical academic accommodations provided through schools' disability support services may not address all their needs (Anderson, 2018; Anderson et al., 2017; Davis et al., 2021; Gelbar et al., 2015). Given these concerns, students with ASD are reported to experience higher rates of postsecondary dropout compared to students without disabilities (Bakker et al., 2023).

Autism Support Programs and Information

There is no single definition of an ASP in the literature or in the law. However, previous reviews identified common elements intrinsic to all ASPs (Barnhill, 2016; Nachman et al., 2021; Viezel et al., 2020). In this paper we define an ASP as a program meeting one or more of the following characteristics: (a) advertising or promoting the program as an ASP; (b) provision of services to students specifically diagnosed with an ASD; (c) services and supports tailored to the needs of autistic students as opposed to students with other disabilities (e.g., intellectual disabilities, learning disabilities) or disabilities in general. With this definition we differentiate ASPs from more general disabilities services offices or programs, but also recognize that an ASP can exist within a disabilities services office. The critical conditions are identifying itself as an ASP and tailoring services specifically to the needs of autistic students.

In addition to academic support, ASPs commonly include social support such as social skills training and specialized transition events, as well as mental health services and peer mentoring programs that are specific to the needs of this population (Viezel et al., 2020). However, it is not clear that these ASD-focused supports exist in all institutions of higher education. While all U.S. postsecondary institutions are required to offer accommodations for students with disabilities under the Americans with Disabilities Act of 1990 (ADA) and Sect. 504 of the Rehabilitation Act of 1973 (King & Piotrowski, 2021), the degree to which supports are tailored to the unique needs of students with ASD cannot be determined easily. Therefore, prospective college students with ASD who need such support must locate universities that offer them, but the process is opaque.

Challenges in Accessing Information About ASPs

Previous research has identified a combined 100 ASPs at 2- and 4-year colleges and universities in the US, representing only 3% of total schools. (Barnhill, 2016; Nachman et al., 2021; Viezel et al., 2020; Wischnewsky, 2023). This means that prospective students, their families, and other advocates must identify which handful of institutions have these programs as they consider where to submit applications to 2-year or 4-year institutions. Although parents of autistic children may consult social media to determine which ASPs are best suited to their child's needs based on others' experiences (Pickard & Ingersoll, 2015), this information may not accurately represent services offered at a particular institution. Therefore, potential applicants, their families, and other advocates must engage in manual internet searches to determine which postsecondary institutions offer services targeted specifically at meeting the needs of autistic students.

Some online resources which aim to collect information about ASPs in a single repository are available ("ASPs", 2021; "College programs", 2023). While such a resource may prove useful to prospective students, their families, and other advocates there is reason to doubt their adequacy and thoroughness. As seen in previous research on this topic, similar efforts to collect data about ASPs can lead to varying results (Nachman et al., 2021; Viezel et al., 2020). Further, in our research we found that the nature and existence of an ASP at a particular institution can change over time. Thus, a static resource which is not regularly updated will soon be inaccurate and/or incomplete. To make such regular updates practical, some degree of automation is needed.

Moreover, given the complex support needs intrinsic to this population (Cai & Richdale, 2016; Jackson et al., 2018), prospective students and their families will require specific information about services offered and how to access them. For example, do programs require documentation of

an ASD diagnosis? What documentation is accepted? Do they provide services to address secondary diagnoses that are more likely to occur with ASD? Are there extra fees or other contingencies for participation? Do the services offered align with the social, academic, mental health and career needs of a prospective student? Students may also want to know how effective these programs are for service recipients in supporting degree attainment, and what other outcomes they produce (e.g., completion certificates). Here, again, parents and potential students must rely on manual internet searches, potentially followed by direct inquiries to ASP staff, to glean this information.

Knowledge Gaps in ASP Services Offered

Previous research has investigated the occurrence and nature of ASPs to answer questions about the nature of ASP services (Barnhill, 2016; Nachman et al., 2021; Viezel et al., 2020). Barnhill (2016) conducted an internet search for ASPs, as well as a literature review which identified several postsecondary institutions which offered specific services for students with Asperger's syndrome and ASD. They then gathered information about the identified ASPs by contacting university officials to fill out a survey. Through this search procedure they identified 45 schools, of which 30 completed their survey. They did not report the names of the institutions that were identified, and thus the information was not immediately applicable to prospective students. They did report the types of support offered to students; however, they did not report on potential barriers to access the programs such as application process, diagnostic criteria or program fees. Additionally, they did not report on the programs' administrative housing. Viezel et al. (2020) utilized the Google search engine, entering terms from a list of queries such as "autism college program" or "college autism support" along with the name of each U.S. state. They then manually reviewed the first four pages of results, considering only pages that linked to an official institution website. This search procedure identified 55 schools with ASPs, which were specifically reported in the paper. They reviewed the webpages of the identified ASPs to gather information about the programs and surveyed program staff to fill in missing data. They reported on admission criteria, program cost, support offered and administrative housing. While their inclusion criteria were limited to 4-year degree granting institutions, it is unclear if each program specifically served degree seeking students. Nachman et al. (2021) used the Carnegie Classification of Institutions to identify all degree granting institutions in the U.S. and manually reviewed each institution's webpage, by entering the term "autism program" into the search function of the website's homepage. They identified 74 ASPs and reported institution type and geographic region but did not report information on admission criteria, administrative housing, program cost, supports offered, or degree status of the programs. However, their results were incorporated into a regularly updated list of ASPs at the College Autism Network website (http://can.softr.app). At the time of writing, this website hosts information on 94 ASPs.

Need for the Current Study

The existing body of research and related resources has identified a combined 100 ASPs at postsecondary institutions in the U.S. According to the National Center for Education Statistics, there are nearly 4,000 degree granting postsecondary institutions in the U.S. Given the overall number of institutions, finding the apparent few that offer ASPs is a monumental task. Previous research relying on labor intensive, manual internet searches and subjective follow-up surveys to identify ASPs, yielded varying results, and is likely not identifying all ASPs for potential college students. For example, the original publications of Viezel et al (2020) and Nachman (2021) contain 80 identified ASPs, but only 44 were identified by both studies. The lack of overlap in their results suggests that labor intensive manual methods do not accurately identify all postsecondary institutions offering ASPs; existing methods will not yield a comprehensive account of ASP offerings. Further, a manual internet search is likely to be the method by which prospective students and their families discover ASPs. The difficulty of identifying available ASPs and their services through this unreliable and labor-intensive method may be a barrier decreasing the accessibility of higher education for this population.

Additionally, salient gaps in our knowledge about the nature of ASP services offered at U.S. postsecondary institutions remain. Specifically, the total number of institutions hosting ASPs, along with a comprehensive description of their services, is not reported in a single source. Due to the difficulty of discovering these programs and the dynamic nature of their existence, one report which accurately captures the state of ASPs in the U.S. is unlikely to ever exist. However, given limitations of previous searches reported in the literature, there is an urgent need to capture data about the locations of programs, populations served, admission and diagnostic criteria, program cost, administrative housing, supports offered, institution type and degree granting status of each identified program. No previous work has included all these data points in their results.

The purpose of the present study is two-fold. First, we evaluate a new web-search procedure for ASPs, based on established techniques from data science (Grishman, 2015;

Le & Mikolov, 2014), such as information retrieval and text classification. We assess whether our use of automation increases efficiency of search compared to previous methods that have been reported in the literature (Barnhill, 2016; Nachman et al., 2021; Viezel et al., 2020). Additionally, we seek to determine whether our procedure is more robust than previous efforts as it results in the most comprehensive account of ASPs to date. Second, given a dearth of reliable information on the characteristics of these programs, we use our comprehensive accounting of ASPs to collect data about the nature of these programs, to answer the following questions: (1) Which post-secondary institutions in the U.S. host ASPs? (2) What are the admissions requirements / costs associated with the programs? (3) What types of support are offered by each program? (4) What type of degree or certificate does the program confer?

Methods

Positionality

This research emerged from an interdisciplinary team of doctoral students and professors in the fields of Computer Science, Special Education, and Organizational Psychology. Each team member brings unique perspectives related to their field of study. Specifically, the teams' combined experience reflects extensive knowledge of information retrieval, data management, accessible information technology and data science, effective supports for postsecondary students with disabilities, including those who identify as autistic, and understanding of organization systems with secondary and postsecondary institutions. Our shared purpose is to increase access to postsecondary educational opportunities for autistic students who may require additional support. This line of inquiry emerged from an identified need in the literature that our team possesses the skills to address.

Search Procedure

Following the direction of previous researchers (Barnhill, 2016; Nachman et al., 2021; Viezel et al., 2020), we chose to conduct an internet search for ASPs. Importantly, the purpose of our search was to replicate a likely internet search of a prospective student, parent, or professional searching for an ASP. Therefore, we did not employ a search process that would be typical of a systematic review (Page et al., 2021), but rather we selected and entered search terms into Google that would be typical of a standard internet search, as was done in previous work (Barnhill, 2016; Nachman et al., 2021; Viezel et al., 2020). This decision operated under the assumption that if a college hosts an ASP, there will be a webpage somewhere on the institution's website that describes the ASP. Additionally, this was a practical decision as it reflects the method that a prospective student and their family are likely to use to search for an ASP. However, by leveraging various data techniques, we automated a large portion of the search and discovery process to reduce the amount of manual labor required. We programmatically issued search queries to quickly gather a large set of webpages. We then utilized vector representations of the documents to classify them, reducing the number that must be manually examined (Fig. 1). These techniques are described in more detail below.

The goal of our internet search is to discover at least one webpage from each ASP hosting institution. There are close to 4,000 degree granting institutions in the U.S. An ideal search engine with an optimal query would thus provide all webpages that we seek within the first 4,000 results. However, no such search engine exists. Additionally, many institutions which host an ASP will have multiple webpages about that ASP, thus the search engine will return multiple results from the same institution increasing the total number of results required to capture all institutions. Given these limitations of existing search engines, we utilize advanced search features to enhance our query results.

We start by consulting with research team members who have relevant domain knowledge to modify and expand the query list of Viezel et. al (2020). This gives us the following query pattern *autism + college + program OR college + autism + support OR ASD + college + program OR ASD + college + support OR autism + university + program OR autism + university + support OR ASD + university + program OR ASD + university + support*. Search engines have advanced features, which allow one to fine-tune their query and search space. For instance, *site:domain.edu* tells a search engine to limit results to the specified domain. We use this feature to send our query for each domain name mentioned in the work by Viezel et. al, (2020) and recorded the number of results before the first occurrence of an ASP

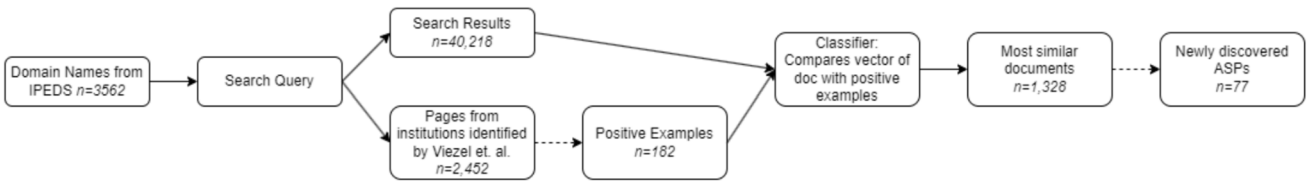


Fig. 1 Search and classification pipeline. Solid lines represent automated steps, dotted lines represent steps that require manual review

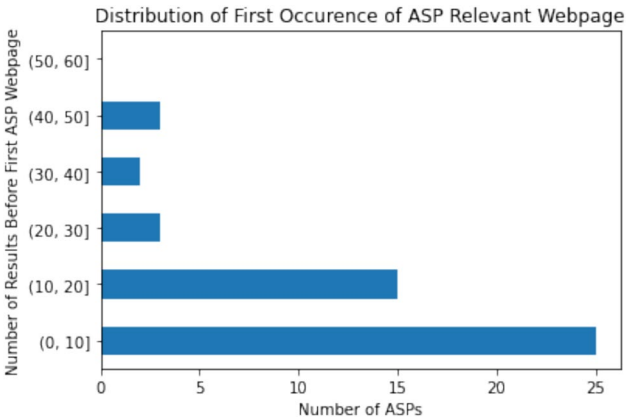


Fig. 2 Distribution of the number of query results before the first ASP relevant web page for a given institution

related webpage. Shown in Fig. 2, we notice that the ASP relevant webpages are most often contained within the first 20 results and never occur outside the top 50 results, therefore we limit our queries to 50 results to reduce the amount of computation required.

To capture all postsecondary institutions, we issue this query with the domain name of each postsecondary institution in the U.S. We utilize the IPEDS database, maintained by the National Center for Education Statistics, to obtain domain names from all degree granting institutions in the U.S (National Center for Education Statistics, n.d.). IPEDS contains data from 4,250 degree granting institutions with 3,562 unique domain names. The discrepancy between these numbers is because many institutions have satellite or affiliate campuses which are recorded separately in the database but utilize the domain name of the parent institution. Given that our search procedure relies on domain name, we utilize only the domain of the parent institution. This contains n = 3,562 domain names, of which 771 are from public 2-year institutions, 751 are from public 4-year institutions, 269 are from private 2-year institutions and 1771 are from private 4-year institutions.

To test the quality of our query, we experiment with query expansion. Query expansion can be achieved using sequence to sequence models which generate potential queries given a set of documents (Nogueira et. al, 2019). Search engines such as Google provide "Related searches," which one can use as a form of query expansion. Related queries appear as suggestions at the bottom of the search results page in Google. For example, for the query *autism+college+program*, Google will give the related queries in Fig. 3. Under the assumption that Google uses state of the art techniques to suggest queries, in this work, we use its related queries for experimentation. However, we take steps to keep our utilization of the search engine *polite* (Najork, 2009). That is, we keep track of the list of related queries and avoid submitting duplicate queries. For example, the query ``online college for autistic students'' appears in the list of related queries of both *autism+college+program* and *college+autism+support*, but we only issue it once. Our experimentation with this form of query expansion yielded little results, that is issuing related queries did not lead to a significant increase in the number of webpages captured. This suggests that our expert developed query list was well developed and sufficient for this particular task.

We issued our expert developed query, with the pattern *site:domain.*edu for each domain found in the IPEDS database for a total of 3562 total queries. We recorded up to the top 50 results for each query (some queries yielded fewer than 50 total pages). After completing the search procedure, the resulting data set included 55,291 web pages. Of these pages, 15,073 were of a file type such as.pdf or.doc or a 404-error page. For ease of processing the data, we excluded non-html pages. Using the python library beautifulSoup4, we extracted the text content of each webpage.

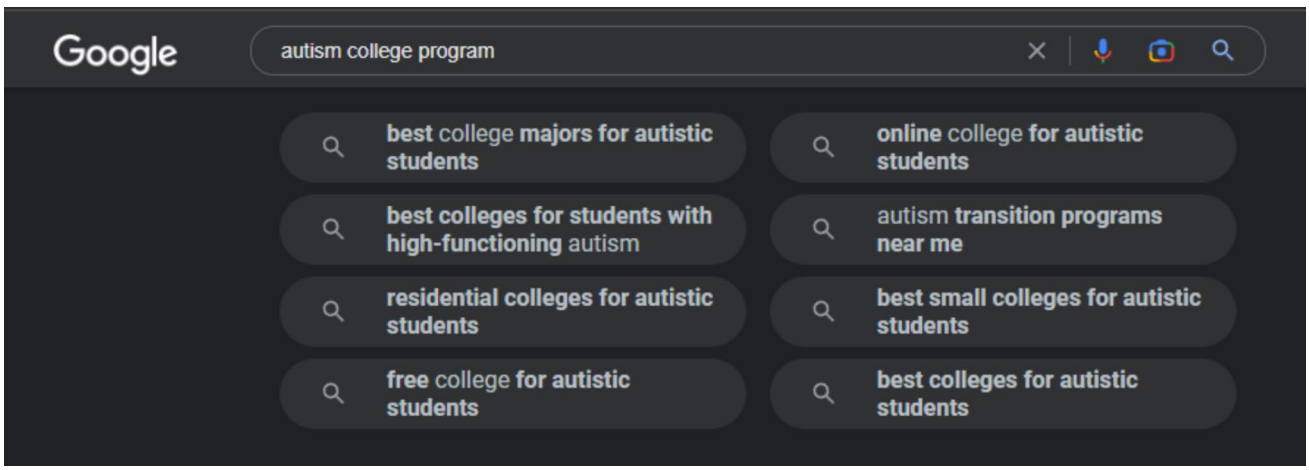


Fig. 3 An example of Google's related queries

The extracted text feeds into the next component of our pipeline: classification.

Classification

The resulting dataset of 40,218 webpages is far too large to manually evaluate. Our goal is to reduce this dataset to a size that is reasonable for manual evaluation by selecting the most relevant documents. Using the contribution by (Viezel et. al., 2020) we first manually evaluated the documents that our search procedure returns from the 55 institutions that they identified as hosting ASPs. There were 2452 pages from those 55 schools. After manual evaluation, 182 of those webpages were identified as directly related to the ASP hosted at the parent institution. These webpages are used as examples of positive instances which can be used to train our classifier. The documents from institutions identified by (Nachman et. al., 2021) are set aside as a test set to evaluate the accuracy of the classifier.

After manually labeling these documents, we trained a *doc2vec* model on the entire corpus of webpages, treating each webpage as an individual document. *doc2vec* is a neural network-based model which utilizes a form of unsupervised learning called continuous bag of words to learn fixed length vector representations for documents (Le & Mikolov, 2014). We chose *doc2vec* for this problem instance over similar representations, such as tf-idf, because the document level vector representations are able to capture the semantic meaning of a document and result in dense vectors (Minaee et al., 2021). The semantic content of the documents is advantageous, as it is what we are trying to capture. The density of the vector is necessary to allow similarity comparison to be computationally tractable. Additionally, *doc2vec* is generally better suited for a dataset that contains documents of variable length, which is the case in our data. Once the *doc2vec* model is trained, we compute vector representations of each document in our corpus and treat them as features for classification.

With our document vector representations, we proceed under the assumption that documents which refer to ASPs will be semantically similar to each other. Given that *doc2vec* generated vector representations capture semantic content, we further assume that the vectors will also be similar. Therefore, we can discover semantically similar documents by comparing the vector representations of the documents. Utilizing the positively labeled data that we obtain from the subset of documents from the institutions identified by (Viezel et. al., 2020), we attempt to discover the documents in the corpus which are most like these positively labeled examples. We accomplish this by computing cosine similarity of the vector representations between each positively labeled document and all other documents in the corpus. Cosine similarity is a measure of the similarity of vectors and is often used to compare document similarity in problems akin to this one. The top 20 most similar documents are recorded for each positively labeled document. This gives a sub-set of 1,328 documents.

Data Collection and Analysis

With this set of 1,328 documents, we proceed with manual data analysis. Each of these documents corresponds to a webpage on a university website. A primary researcher reviewed all 1.328 webpages to determine if they met our inclusion criteria. Webpages for which inclusion was ambiguous or uncertain were flagged. The primary researcher then met with a second researcher to review these flagged cases. If the two researchers did not agree, the page was reviewed in a group session by the entire research team to reach a final consensus decision. For inclusion in the study, the webpage must describe a program which meets one of the three criteria we provide as definition for an ASP: (a) advertising or promoting the program as an ASP; (b) provision of services to students specifically diagnosed with an ASD; (c) services and supports tailored to the needs of autistic students as opposed to students with other disabilities (e.g., intellectual disabilities, learning disabilities) or disabilities in general. For example, a webpage which provides information for faculty about accommodating autistic students would not be included. Through this manual review, we discovered 99 new documents which directly reference ASPs. A number of these documents reference the same ASP, yielding a total of 77 new ASPs which were not identified by the previous work.

To test the robustness of our procedure, we then compare our results to the data hosted on can.softr.app, which is an updated list maintained by the authors of (Nachman et. al., 2021). Of the 94 institutions identified in that data, 48 of them were also included in the work by (Viezel et. al., 2020). Of the remaining 46 institutions, our procedure identified 28 of them. In addition, our procedure identified 49 institutions that were not previously identified in the literature, which represents a near 50% increase in the total number of ASPs reported in the literature.

A key question is whether the 49 newly identified programs represent false negatives from prior studies—that is, whether they existed at the time but were missed due to limitations in earlier search methods. However, determining the start date or online presence of an Academic Support Program (ASP) is not straightforward. To address this, we leverage the Internet Archive (web.archive.org), which timestamps archived web content. For each of the 49 institutions not previously reported in the literature, we searched

the Internet Archive for the earliest available version of their ASP websites.

We found archived versions for 38 of the 49 institutions. We then compared the earliest archived date of each site to September 2017, the data collection start date in the earliest known ASP study (Viezel et al., 2020). Of these, 25 websites appeared after September 2017, indicating newly emerged programs. The remaining 13 had websites dated before September 2017 and were likely missed by previous search procedures.

These findings highlight two important features of our automated discovery pipeline: (1) its ability to detect programs overlooked by earlier methods, and (2) the dynamic nature of ASPs, with new initiatives continually emerging—underscoring the need for ongoing updates to any comprehensive ASP database.

Our next step is to gather detailed information about each program in our dataset. To gather information about each ASP we manually examined the webpages of each identified institution. Combining ASPs identified in previous works and those identified by our procedure, we evaluated webpages from 149 institutions. During data collection, 16 ASPs reported in previous works were found to no longer be active, resulting in 133 ASPs. This further emphasizes the evolving nature of ASPs and a need for an automated solution to the problem that updates the data periodically with minimal human effort. The data collection process consisted of visiting the home webpage of the ASP and following any link to a "downstream" related webpage. Research assistants manually evaluated each webpage for specific information about the characteristics of the program. In total, this portion of data collection required 14 h of data extraction from research assistants. Collected information included the population the program is targeted to serve, whether the program requires a program specific application, elements of the application if one exists, diagnostic requirements for participation, program cost, administrative home of the program, specific supports offered, duration of support offered, hours per week of support offered, number of students enrolled in the program, number of full-time staff employed, number of part-time staff employed and whether students enrolled in the program are eligible to earn a degree or a certificate of completion.

Table 1 Supports offered by ASPs

Support	Number of Programs which Include
Academic Coaching	100 (74%)
Peer Mentoring	84 (62%)
Vocational Support	78 (58%)
Social Events	69 (51%)
Mentor Check-Ins	60 (44%)
Social Skills Interventions	50 (37%)
Designated Courses	45 (33%)
Assistance Communicating with Campus Staff	40 (30%)
Independent Living Support	37 (27%)
Tutoring	36 (26%)
Pre-Semester Orientation/Transition Program	34 (25%)
Internship	33 (24%)
Life Coaching	30 (22%)
Study Hall	26 (19%)
Mental Health Services	26 (19%)
Job Placement	19 (14%)
Parental Support	15 (11%)
Weekly Workshops	14 (10%)
Physical Fitness	13 (10%)
Peer Group Meetings	11 (8%)
Relationship / Gender Identity / Sexuality Support	10 (7%)
Group Counseling	10 (7%)
Designated Housing	10 (7%)
Testing Accommodations	6 (4%)
On-Call Support	5 (4%)
Speech / Language Intervention	4 (3%)
Sensory Reduced Spaces	4 (3%)
Assistive Technology	3 (2%)
Priority Registration for Courses	2 (1%)

Results

Using information from previous research and a new search procedure, we identified 133 active ASPs at degree granting institutions in the U.S. Further, 49 of these institutions were newly identified by this work. A list of identified programs, as well as select information about each program is provided in Table 1. It is important to note that these data were gathered in April 2023 and may change over time. Further, the information reported here is reliant on what is provided on the ASP's website; it may be the case that some requirements or services of an ASP exist but are simply not present on the website. However, this information reflects what is available to prospective students and their families as they weigh the postsecondary educational opportunities available to them. Summary statistics about the programs in aggregate are given to show the state of affairs for various stakeholders in this space such as parents, prospective students, university administrators, policy makers and researchers.

Population Served

The target population was determined by researchers based on the language used on the websites of the programs. For example, one program stated: "This program is designed to meet the unique needs of students with Autism Spectrum

Disorder". A majority of programs specifically target students with an Autism Spectrum Disorder, including Asperger's Syndrome and Pervasive Developmental Disability – Not Otherwise Specified (55.6%). Fewer programs are designed for students with Intellectual and/or Developmental Disabilities (IDD) (15.0%), for Neurodiversity generally (9.8%), for students with "Learning Disabilities" (3.8%) or for students with Attention-Deficit/Hyperactivity Disorder (0.8%). Some programs specified that they serve students with a variety of conditions such as Autism and/or IDD (5.3%), Autism and/or Neurodiversity (6.0%), and Autism and/or Learning Disabilities (0.8%). The remaining programs (3.2%) did not specify a clinical diagnosis as part of their target population.

Application Process

Most of the programs (74.4%) require a program-specific application for admission. In the case of programs which serve degree seeking students, the program application is distinct from the application for admission to the college or university and requires applicants to be admitted to the school before they will be considered for admission to the ASP. Information about a program application is not readily available for 17.3% of programs and the remaining 8.3% do not require an application. Information about the elements required for the application is only readily available for 51% of the programs which require an application. For example, if the application was a multi-part online form, the researchers did not complete the form to avoid creating false applications that program staff would have to process. Of the programs that have a readily available application, 98% require an interview with the student, 94% require a High School transcript, 64% require a writing sample, 58% require letters of recommendation, and 15% require standardized test scores.

In addition to a program application, most programs (92.5%) require some proof of diagnosis. Information about diagnostic criteria was readily available for 64% of programs that require proof of diagnosis. Of those programs, 53% require a High School IEP (Individualized Education Plan), 51% require an ASD diagnosis, 50% require a Psychological Evaluation conducted outside the college or University, 6.25% require a Psychological Evaluation conducted at the college or University, 8.75% only require that participants self-identify as neurodiverse, 7.5% require a diagnosis of IDD and 3.7% require a diagnosis of a "Learning Disability." Additionally, 17.5% of programs require that Psychological Evaluations include IQ and Achievement testing such as the Stanford-Binet and Woodcock-Johnson. It is common for programs to require multiple diagnostic elements as part of the program application.

Cost

Information about the cost of participation in the program was available for 72% of programs. Of these programs, 22% are offered at no additional cost to students. The remaining programs charge students an additional fee per semester in addition to tuition (for matriculated students). The per semester fee for program participation ranges from $75 to $19,640 with a mean of $4603, median of $3670 and standard deviation of $3264. Figure 4 shows a distribution of the costs of programs. Figure 5 shows the comparison and correlation between the cost of ASPs and the median income of their states, separated by public and private institutions. The distribution shows that, in absolute terms and relative to median state income, the cost of ASPs is quite expensive, especially when considering that ASP cost is most often in addition to the cost of undergraduate tuition. However, some programs are not degree granting, and therefore do not charge tuition; these programs tend to have a higher ASP cost, as is the case with the most expensive program shown in the distribution. This potential financial burden has been ameliorated through the Higher Education Opportunity Act of 2008, in which the federal government created a new classification for these non-degree programs. This classification allows students with intellectual disability to access federal financial aid. However, it is unclear how many institutions have sought or obtained the necessary classification from the government to allow their students access to this funding, as this distinction is not recorded in the IPEDS database.

Administrative Home of the Program

Organizational structure divides the work of the organization, differentiates, and coordinates the work, and then integrates the work to meet the organization's mission and goals. Organizational structures of a college or university affect student learning and engagement (Berger, 2002; Lambert et al., 2007; Porter, 2006; Rowan-Kenyon, 2009). These effects extend beyond physical proximity to the interactions and the culture of any given sub-unit (Brunton, 2018). In the case of ASPs, one that is in a disabilities services office may be better coordinated with the larger institutional initiatives in disabilities accommodations, compared to one that is in a research laboratory or academic department, which may be more aligned with research or academic concerns. As is the case with other departments and functions in a college or university, there is no "one best way" to organize or structure an ASP. The ideal structure is one that best serves the needs and mission of a particular institution. ASPs are expected to reside in various administrative homes, as organizational

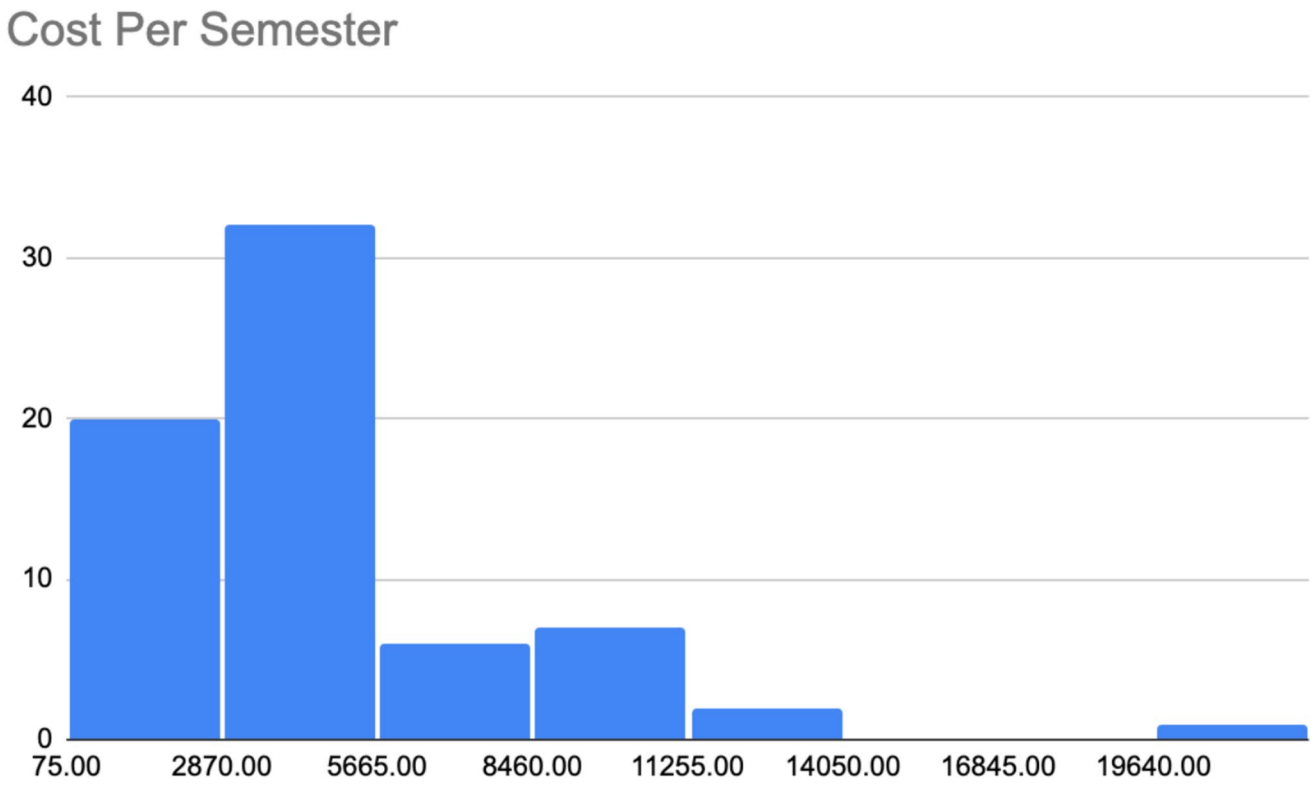


Fig. 4 Distribution of the cost per semester of program participation

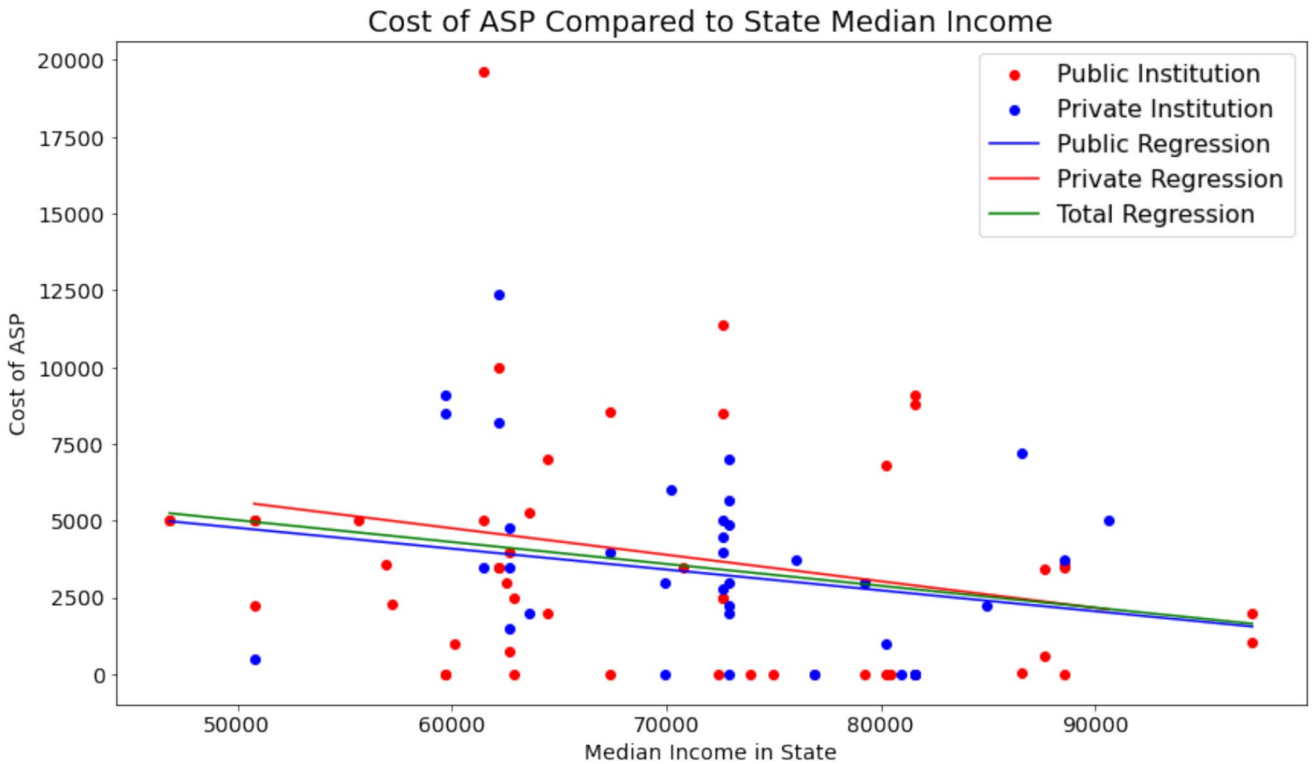


Fig. 5 Cost of participation in an ASP compared to the median household income of the state of the institution

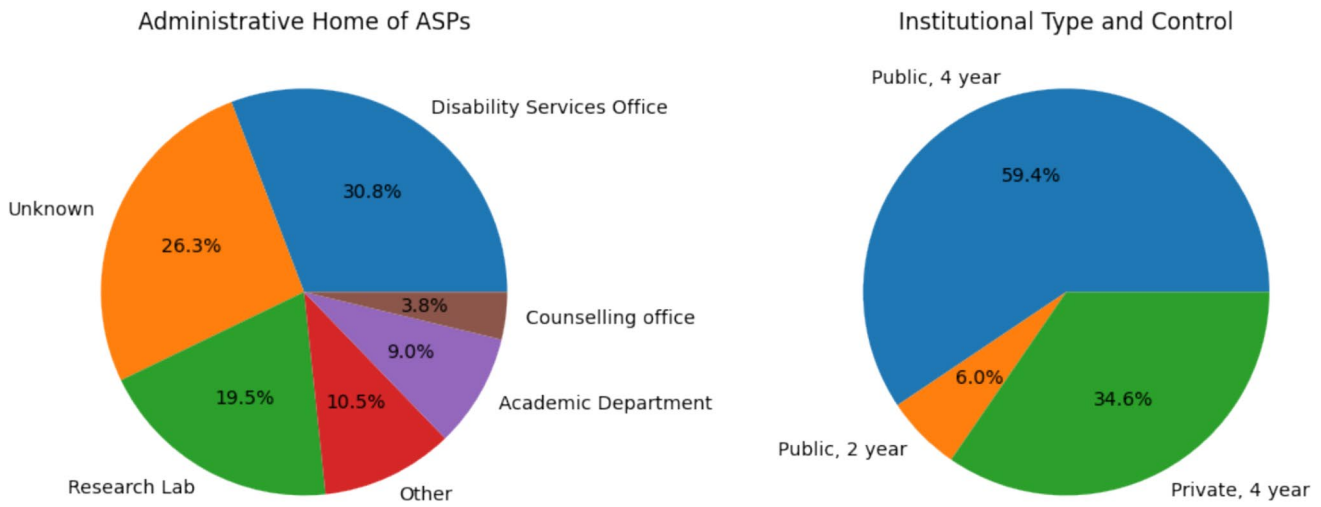


Fig. 6 (Left) Breakdown of administrative homes of ASPs. (Right) Breakdown of Institution Type (2 year or 4 year) and Institutional Control (Public or Private)

structure is one of many ways organizations adapt to their ever-changing environments (Weick, 1974).

The administrative housing of ASPs, shown in Fig. 6, varies between institutions. The most common housing is the university's Disability Services Office (30.8%). It is important to note that this office provides accommodations required by the Americans with Disabilities Act (ADA), however for inclusion in this paper a program must provide accommodations above those required by the ADA to include those that are specific to meeting the needs of students with ASD. As with much of the data we sought to collect, the administrative home was not clearly stated on the webpage for many of the ASPs that were included (26.3%). The next most common administrative home is a Research Lab, Center or Clinic (19.5%) or an Academic Department (9.0%). The least common administrative home is the university's Counseling or Psychological Services Office (3.8%). A small number of programs (10.5%) were housed in other, less defined administrative offices such as the office of the provost, department of academic affairs or a dedicated neurodiversity initiative. Administrative location may also have implications for budgeting, access, and perhaps longevity of an ASP. An ASP housed in a disabilities office may signal a more institutionalized, and perhaps better resourced, program than one housed in a research lab which may imply a newly established, or "soft money" funded program that is temporary, or perhaps recently developed. An ASP in an academic department or college may offer services only to those enrolled in that department or college.

Institution Type and Control

Figure 6 shows the institutional type and control of postsecondary institutions which host ASPs. The most common is public, 4-year institutions at 59% of ASP hosts, followed by private, 4 years at 35%. Very few ASPs (6%) are hosted at 2-year institutions. This is out of proportion with the total number of institutions, as public, 4-year institutions account for only 21% of all colleges and universities but host most ASPs. Private, 4-year institutions are the largest group at 49% of all colleges and universities but host a smaller proportion of ASPs at 35%. There is a severe gap in service at 2-year institutions, which account for a total of 30% of all colleges and universities but host only 6% of ASPs.

Table 2 Missing Data per Variable

Variable	Percent Missing
Population served	0%
Application required	17.3%
Elements of application	36.6%
Diagnostic criteria of application	33.6%
Cost	27.6%
Duration of support	44.4%
Administrative home	25.8%
Certificate of completion if no degree offered	9.7%

Supports Offered

Table 2 shows the types of support offered to participants of ASPs. As reported in (Viezel 2020), academic coaching and peer mentoring are the most common supports offered. Academic coaching is typically administered by program staff in the form of regularly scheduled meetings to discuss academic progress, course selection, and any challenges the student is facing. Peer mentors are commonly selected from the undergraduate population of psychology and/or education departments, on an employment or volunteer basis, and act as a social conduit for the student. Peer mentor programs typically involve an informal weekly meeting for

the student and mentor to discuss academics, social life and transition to college life. These programs sometimes include tutoring, mentors attending class with the student, group student-mentor social events, or participation in outside student clubs and organizations. The majority of programs also include some form of vocational support, such as resume writing or interview preparation, and recurring social events for program participants.

Other commonly offered supports include regular mentor check-ins (distinct from peer mentor meetings), social skills interventions, designated courses, assistance communicating with campus staff, independent living support, tutoring services, and a pre-semester orientation/transition program. Offerings of other support listed in Table 2 varied widely between institutions; our study only includes supports that are listed on the program's webpage as being offered through the ASP. Some supports may be available to students through other campus programs, but not included here such as tutoring offered through an academic support office, counselling being offered through a psychological services office or testing accommodation offered through a disability services office.

Additionally, we gathered information regarding the amount of support in hours available to students per week. However, most programs (85.3%) did not report this information on their webpage, therefore the data are not reported here. However, a substantial number of programs (41%) did report the number of semesters that support is offered to students. A distribution of duration of support in semesters is shown in Fig. 7. Again, data was not recorded unless explicitly stated on the program's webpage. In cases where the webpage stated that support is offered for the duration of the student's time at the institution, it was recorded as eight semesters for four-year institutions and four semesters for two-year institutions.

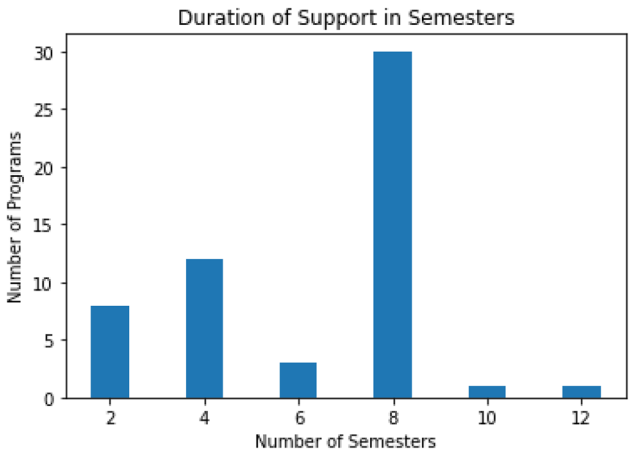


Fig. 7 Duration of support offered by ASP in number of semesters

Degree Offerings

The degree offerings of the identified programs highlight distinct categories of programs that this work and previous works have identified. Most of the identified programs (79.7%) cater to matriculated students who are eligible to earn a bachelor's or associate degree depending on the institution. These programs require the student to first be admitted to the institution in accordance with the admission criteria that all students are subject to before they are eligible to apply to the ASP. The remaining programs, often denoted as Comprehensive Transition Programs (CTP), do not require the student to be admitted to the institution; however, they offer a certificate of completion rather than a traditional degree. We found that CTPs are more often designed for students with IDD rather than ASD.

Missing Data

For all fields of the collected data missing data was a persistent issue. Table 3 shows the percentage of missing data for each variable reported above. In addition to the summary statistics reported above, researchers collected information about the number of enrolled students, number of full and part-time staff employed by the program, and amount of support offered in hours per week. All of these variables were available for less than a quarter of the identified programs and are therefore omitted here. While the absence of such information on the program websites does not necessarily imply that these things do not exist as part of the program, it is representative of the great variability between institutions in the reporting of information about ASPs. This is problematic not only for this study, but also for prospective students and their parents who may struggle to discern which institution is best for them due to the lack of available information.

Discussion

The present work expands on existing work by Viezel et al (2020) and Nachman et al (2021) by identifying Autism Support Programs at postsecondary institutions in the U.S., as well as providing additional contextual information. We introduce a novel search procedure, based on established techniques from the field of data science, which allows us to automate the search and discovery process of ASPs, significantly reducing the amount of time and labor required to conduct this work. The increase in efficiency of this process is important because the expanding and dynamic nature of ASPs necessitates regularly repeated efforts to aggregate data about the programs. Additionally, our search procedure

proved to be robust, identifying 49 programs not previously found in the literature.

Having identified a comprehensive list of ASPs in the U.S., we conducted a manual evaluation of the program's webpages for data collection. These data can provide important insights into the rapidly changing landscape of postsecondary education for adults with ASD and for various stakeholders, such as students, their families, advocates, college counselors, and researchers. Our data collection process indicates a prevalent lack of available information contained on the webpages of ASPs, which is likely to prove problematic for all stakeholders. Given missing data about these known variables, it is not unreasonable to conclude that ASP websites may be missing information about unknown variables, including the types of services offered. This finding underscores the need for institutions with ASPs to provide complete information on their websites so that potential students can make informed choices about whether the ASP will meet their needs.

The following section commences by highlighting the major contributions of our work. Subsequently, we revisit the methodology used for data collection and analysis. Next, we discuss the results of data analysis regarding the characteristics of ASPs, drawing comparisons to the findings of relevant prior works. Finally, we highlight the significant issues of discoverability and missing data, provide recommendations for improvements, and conclude with the limitations of our study.

Major Contributions

The first major contribution of the present work is an improvement in the search procedure employed to gather information about ASPs in the U.S. By approaching this challenge as more than a simple search problem that may be solved by intensive labor, the current study employs data science techniques to improve upon existing search procedures. We argue that our search procedure is more robust than existing efforts. This is shown through testing against existing results and by obtaining new results. Our procedure identifies 49 ASPs that have not been previously reported in the literature. This represents a near 50% increase in the total number of ASPs identified in the literature. Additionally, we leverage data science techniques to automate substantial portions of the procedure, such that our search methods require little manual effort from researchers. This vastly improves the efficiency of conducting a search, making it much more practical to repeat regularly and avoids the problems of leaving information untouched. Having developed an improved search procedure, we aim to use our methods to help solve the accessibility problem for this population.

The second major contribution of the present work is the most comprehensive overview of the characteristics of ASPs in the literature to date. We report on the population served, admission process, diagnostic criteria, program cost, administrative housing, institution type, supports offered, and degree offering of all programs for which information is available. However, the issue of missing information, also noted in the work of Viezel et al (2020), remains prevalent and limits the conclusions that can be drawn about the nature of these programs generally. Missing information is also likely to be a significant issue for other stakeholders such as prospective students, their families, and college counselors. As shown in Table 3, there is a wide range of variability in the reporting of information about ASPs. It is reasonable to assume that the difficulties the researchers had discovering this information would also be experienced by other stakeholders.

Methodology

Our methodology for searching and discovering ASPs shows two significant improvements over previous work. First, the results are more robust. Our search procedure identified 49 ASPs which have previously not been found in the literature. Second, our procedure was mostly automated and required minimal manual labor from researchers. Due to the relatively new nature of ASPs and the ongoing nature of research into best practices for ASPs, the number and characteristics of ASPs are likely to change in the future. As we found in our data collection procedure and search of the web archive, some ASPs which previously existed are now defunct and a number of new ASPs have emerged in the time since previous works. This rate of change necessitates annual efforts to aggregate this data and requires that such efforts comprehensively evaluate all postsecondary institutions to capture changes to previously identified ASPs. Our approach vastly increases the feasibility of conducting such a search and aggregation effort regularly by significantly limiting the amount of manual effort that is required. The data we have gathered and annotated here can also serve as training data for future improvements to our pipeline. Future work may further increase efficiency by employing Information Extraction (Grisham, 2015; Small & Medsker, 2014) techniques to automate the time-consuming data collection portion of our procedure.

Table 3 Characteristics of ASPs

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.adelphi.edu/bridges/	Neuro-diversity Generally	Unknown	Unknown	Self-Identification	$4850	N/A	Unknown	Academic Coaching, Vocational Support, Job Placement Services, Social Skills Interventions, Life Coaching, Tutoring	Yes, 4 Year / Bachelor's Degree
https://www.american.edu/provost/academic-access/learning-services-program.cfm	Learning Disabilities	Yes	High School Transcript, Writing Sample	"Documentation of Learning Disability" https://www.american.edu/provost/academic-access/upload/documentationguidelines_learningdisabilities-2.pdf	$5000	2	Disability Services Office	Peer Mentoring, Tutoring, Special Courses	Yes, 4 Year / Bachelor's Degree
https://sdap.appstate.edu/	IDD	Yes	High School Transcript, Writing Sample, Resume, References,	High school IEP, Psychological Evaluation outside the University, Documentation of ID	$2500	8	Unknown	Academic Coaching, Peer Mentoring, Vocational Support, Job Placement Services, Physical Fitness	No
https://www.apsu.edu/full-spectrum-learning/	ASD	Yes	Interview, Writing Sample	ASD Diagnosis	$3500	8	Unknown	Academic Coaching, Peer Mentoring, Vocational Support, Mentor Check-Ins, Study Hall, Life Coaching, Tutoring, Special Courses	Yes, 4 Year / Bachelor's Degree
https://salt.arizona.edu/about	mild to moderate learning and attention challenges	Yes	High School Transcript, Writing Sample, Application Fee	High school IEP, Psychological Evaluation outside the University, "Most Current Comprehensive Testing" or a "Reflection Prompt" if documentation is not available	$3500	8	Unknown	Academic Coaching, Peer Mentoring, Social Events, Psychological / Mental Health Services, Tutoring, Educational Technology, Workshops	Yes, 4 Year / Bachelor's Degree
https://www.astate.edu/college/education/howltp/eligibility/	ASD, ID	Yes	High School Transcript, Interview	ASD Diagnosis, ID diagnosis	$5000	4	Unknown	Academic Coaching, Peer Mentoring, Vocational Support, Social Skills Interventions, Independent Living Support	4 year degree for ASD students

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://aurora.edu/pathways/pathways-collegiate-program/index.html	ASD	Yes	Unknown	Unknown	$3000	N/A	Unknown	Academic Coaching, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Events, Individual Counselling, Study Hall, Life Coaching, Sensory Support Spaces	Yes, 4 Year / Bachelor's Degree
https://www.beaconcollege.edu/admissions/navigator-prep/	Neurodiversity Generally	Yes	High School Transcript, Unknown, Unknown if more than transcript is required	None		2	Unknown	Parental Support, Life Coaching, Transition Counseling	No
https://www.bellevuecollege.edu/autismspectrumnavigators/	Neurodiversity Generally	No	No Application Required	Self-Identification		N/A	Other	Peer Mentoring, Vocational Support, Parental Support, Special Courses, Facilitated communication with faculty	Yes, 4 Year / Bachelor's Degree
https://dsp.berkeley.edu/support-services/autism-spectrum-services	ASD	Unknown	Unknown	Unknown	0	N/A	Disability Services Office	Academic Coaching, Vocational Support, Social Events, Study Hall	Yes, 4 Year / Bachelor's Degree
https://www.bpcc.edu/pse	ASD, ID	Yes	Unknown	ASD Diagnosis, ID Diagnosis	$2304	N/A	Unknown	Academic Coaching, Vocational Support	No
https://www.bridgewater.edu/life-at-bridgewater/services-for-students/academic-advising-and-support/base-program/	ASD, Neurodiversity Generally	Yes	High School Transcript, Can't see the entire application, also requires parent/guardian to fill out a questionaire	Unknown	$1000	N/A	Unknown	Academic Coaching, Vocational Support, Mentor Check-Ins, Social Events, Psychological / Mental Health Services, Communication with campus / program staff, Independent Living Support, Life Coaching, On Call Support, Transition Supports—early move in, specialized campus tour. Access to sensory reduced study area, guaranteed on-campus employment	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.byui.edu/counseling-center/autism-mentoring	ASD	Unknown	Unknown	Unknown	0	N/A	Counselling or Psychological services office	Peer Mentoring	Yes, 4 Year / Bachelor's Degree
https://www.clemson.edu/academics/studentaccess/autism-transition.html	ASD	Yes	High School Transcript, Interview, Letter of Recommendation, IEP submitted with application but not accepted alone for diagnostic criteria	ASD Diagnosis	$3000	N/A	Disability Services Office	No specific supports listed on website	Yes, 4 Year / Bachelor's Degree
https://www.coastal.edu/scoess/departmentsandprograms/lifeprogram/	IDD	Yes	Interview, Program Application Packet (contents unknown)	significant limitations in intellectual functioning and in adaptive behavior, as defined in AAIDD. (Full Scale IQ Below 70)		8	Academic Department	Vocational Support, Job Placement Services, Independent Living Support, Tutoring, Special Courses, Internships	No
https://reach.cofc.edu/about-reach-program/index.php	IDD	Yes	High School Transcript, Letter of Recommendation, Behavior/Discipline Records, Personal Support Inventory completed by parent, Current IEP	Psychological Evaluation outside the University		8	Unknown	Peer Mentoring, Vocational Support, Social Skills Interventions, Social Events, Independent Living Support, Tutoring, Special Courses, Physical Fitness, Assistive Technology, Internships	No
https://www.chhs.colostate.edu/ccp/programs/belong-program/	Neurodiversity Generally	Yes	Unknown	Unknown		2	Unknown	Independent Living Support	No
https://www.columbusstate.edu/tlc/goals.php	IDD	Yes	Unknown	Unknown		4	Academic Department	Peer Mentoring, Special Courses, Internships	No
https://www.csi.edu/accessibility-services/autism-support-program.aspx	ASD	No	SAS Intake form	Unknown		N/A	Disability Services Office	Academic Coaching, Mentor Check-Ins, Communication with campus / program staff	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.csueastbay.edu/clp/	ASD	Yes	Interview	Unknown	$9089	N/A	Unknown	Academic Coaching, Vocational Support, Job Placement Services, Social Skills Interventions, Social Events, Group Counselling, Independent Living Support, Study Hall, Speech / Language Interventions	Yes, 4 Year / Bachelor's Degree
https://www.cuw.edu/about/offices/ablelight.html	IDD	Yes	Unknown	Unknown		N/A	Unknown	Very vague about the whole program	No
https://www.daemen.edu/student-life/student-services/student-success-center/accessibility-services/college-autism	ASD	Unknown	Unknown	Unknown	$2000	N/A	Disability Services Office	Academic Coaching, Peer Mentoring, Vocational Support, Job Placement Services, Individual Counselling, Psychological / Mental Health Services, Life Coaching, assistance with internship, resume, cover letter and job search	Yes, 4 Year / Bachelor's Degree
https://www.dean.edu/support-success/student-services-resources/learning-support-services/arch-learning-community/	Learning Disabilities	Yes	High School Transcript	High school IEP, Psychological Evaluation outside the University, "Recommend a full-scalle IQ of 90 or above", "Johnson-III-Tests of Cognitive Ability or Stanford Binet IV (preferred)"	$7200	8	Disability Services Office	Academic Coaching, Vocational Support, Psychological / Mental Health Services, Tutoring, Special Courses, Specialized orientation, internship placement	Yes, 4 Year / Bachelor's Degree
http://www.defiance.edu/autism/affinity/program.html	ASD	Yes	Writing Sample, Letter of Recommendation	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University, IQ Test>80, Results from achievement testing (e.g., Wechsler Individual Achievement Test, Woodcock Johnson Tests of Achievement)	$3500	N/A	Research Lab	Academic Coaching, Vocational Support, Job Placement Services, Mentor Check-Ins, Social Skills Interventions, Social Events, Independent Living Support, Study Hall, Tutoring, Relationship / Sexuality / Gender Identity Support	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://drexel.edu/counselingandhealth/autism-support/overview/	Neuro-diversity Generally	Yes	Personal questions about goals	No diagnosis required, IEP encouraged	$2,800	N/A	Counselling or Psychological services office	Academic Coaching, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Events, Study Hall, Special Courses, Summer Transition Program	Yes, 4 Year / Bachelor's Degree
https://dsu.edu/student-life/counseling-center/strong.html	ASD	Unknown	Unknown	Unknown	0	N/A	Counselling or Psychological services office	Seems to be a help to access other on campus supports i.e. disability office, counseling center, etc	Yes, 4 Year / Bachelor's Degree
https://studentaffairs.du.edu/learningeffectiveness	Learning Disabilities	Yes	Unknown	High school IEP, Psychological Evaluation at the University	$2250	N/A	Other	Academic Coaching, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Skills Interventions, Social Events, Life Coaching, Tutoring, Executive Function Coaching, Summer Transition Program	Yes, 4 Year / Bachelor's Degree
https://www.eastern.edu/student-life/counseling-academic-support/college-success-program-students-autism-spectrum-disorder	ASD	Yes	High School Transcript, Interview, Writing Sample	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University, IQ and achievement indicators	$4459	N/A	Counselling or Psychological services office	Academic Coaching, Peer Mentoring, Social Skills Interventions, Social Events, Group Psychological Counselling, Communication with campus / program staff, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree
https://stepp.ecu.edu/about/	Learning Disabilities	Yes	High School Transcript, Standardized Test Scores, Interview, Writing Sample, Letter of Recommendation	High school IEP, Diagnosis of Learning Disability along with IQ and Achievement tests	0	10	Unknown	Peer Mentoring, Vocational Support, Mentor Check-Ins, Independent Living Support, Study Hall, Tutoring, Special Courses, Pre-Semester Orientation / Transition Program, Assistive Technology, Designated Residence Hall	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.edgewood.edu/cutting-edge	IDD	Yes	High School Transcript, Interview, Writing Sample, Letter of Recommendation	High school IEP		12	Unknown	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Job Placement Services, Mentor Check-Ins, Social Events, Independent Living Support, Life Coaching, Special Courses, Physical Fitness, Relationship / Sexuality / Gender Identity Support, seperate Summer Program and Alumni Program	Yes, 4 Year / Bachelor's Degree
https://www.edinboro.edu/academics/support/accessibility-services/basis.php	ASD	Unknown	Unknown	Unknown		N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Events	Yes, 4 Year / Bachelor's Degree
https://www.eiu.edu/step/	ASD	Yes	Interview, Unknown	Unknown	$3000	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Skills Interventions, Social Events, Psychological / Mental Health Services, Communication with campus / program staff, Study Hall, Parental Support, Special Courses, Speech / Language Interventions, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.emich.edu/collegesupportsprogram/	ASD	Yes	Interview	ASD Diagnosis	$7000	N/A	Unknown	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Skills Interventions, Psychological / Mental Health Services, Communication with campus / program staff, Independent Living Support, Study Hall, Tutoring, Physical Fitness, 1:1 Help in Class, Facilitated communication with Faculty (filed under communication with campus/program staff)	Yes, 4 Year / Bachelor's Degree
https://www.esu.edu/cills/index.cfm	IDD	Yes	High School Transcript, Interview, Letter of Recommendation	High school IEP, Must meet eligibility guidelines for having an intellectual disability using the criteria as established by the Pennsylvania State Office of Developmental Programs: IQ< =70	$8487	6	Academic Department	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Social Skills Interventions, Independent Living Support, Tutoring, Physical Fitness, 1:1 Help in Class, Designated Housing program off campus	No
https://www.fdu.edu/academics/colleges-schools/psychology/compass-program/	ASD	Yes	High School Transcript, Standardized Test Scores, Interview	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University	$3740	4	Academic Department	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Individual Psychological Counselling, Group Psychological Counselling, Communication with campus / program staff, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://kremen.fresnostate.edu/centers-projects/wayfinders/index.html	IDD	Unknown	Unknown	Unknown		4	Academic Department	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Mentor Check-Ins, Social Skills Interventions, Social Events, Independent Living Support, Life Coaching, Special Courses, Physical Fitness, 1:1 Help in Class, Designated Dorm	No
https://excel.gatech.edu/home	IDD	Yes	High School Transcript, Interview, Letter of Recommendation	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University, Cognitive Ability, Achievement and Social / Emotional Scale batteries	$19640	8	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Social Skills Interventions, Independent Living Support, Tutoring, Special Courses, Physical Fitness	No
https://coe.georgiasouthern.edu/eagle-academy/	young adults whose disability is characterized by limitations in both intellectual functioning and adaptive behavior	Yes	High School Transcript, Letter of Recommendation	High school IEP, Psychological Evaluation outside the University, IQ test and Achievement Assessment	$5000	4	Academic Department	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Social Skills Interventions, Independent Living Support, Tutoring, Special Courses	No
https://www.gmercyu.edu/academics/programs/integrated-studies-program	IDD	Yes	High School Transcript, Interview, Letter of Recommendation	High school IEP, Documented Diagnosis of ID	$5000	8	Unknown	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Mentor Check-Ins, Social Events, Tutoring, Pre-Semester Orientation / Transition Program	No

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://ds.gmu.edu/masi/	ASD	Yes	High School Transcript, Writing Sample, Letter of Recommendation, Application Fee	Psychological Evaluation outside the University	$6800	N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Skills Interventions, Social Events, Communication with campus / program staff, Special Courses, On Call Support	Yes, 4 Year / Bachelor's Degree
https://www.gvsu.edu/dsr/campus-links-96.htm	ASD	Yes	Interview	ASD Diagnosis		N/A	Disability Services Office	Peer Mentoring, Mentor Check-Ins	Yes, 4 Year / Bachelor's Degree
https://www.harpercollege.edu/ads/programs.php	ASD	Yes	High School Transcript, Standardized Test Scores, Interview, Writing Sample	ASD Diagnosis		N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Job Placement Services, Social Events, Parental Support, Special Courses, Relationship / Sexuality / Gender Identity Support, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree
https://ini.illinois.edu/	Neurodiversity Generally	Yes	Interview	Unknown	$0	N/A	Unknown	Peer Mentoring, Vocational Support, Job Placement Services, Social Skills Interventions, Social Events, Individual Psychological Counselling, Psychological / Mental Health Services, Parental Support, Special Courses, Weekly Mental Health Check In	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.iup.edu/education/centers/labyrinth/	ASD	Yes	Standardized Test Scores	Unknown		8	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Events, Study Hall, Life Coaching, Special Courses	Yes, 4 Year / Bachelor's Degree
https://www.iusb.edu/students/student-support-services/disability-support-services/titans-as-program.html	ASD	Unknown	Unknown	Unknown		N/A	Disability Services Office	Mentor Check-Ins, Social Events, Communication with campus / program staff	Yes, 4 Year / Bachelor's Degree
https://www-s3-live.kent.edu/s3fs-root/s3fs-public/file/Autism%20Advocates%20Program.pdf?VersionId=yGPVtL28ArfB_hySdWaE64CNCFMcfSD4#:~:text=This%20program%20allows%20the%20student,the%20ASD%20student%20in%20mind	ASD	Unknown	Unknown	Unknown		N/A	Unknown	Paired with a faculty member, meet with them 3 times per semester	Yes, 4 Year / Bachelor's Degree
https://www.kirkwood.edu/explore/services/learning-services/ask-program	ASD	Yes	Unknown	Unknown	$0	N/A	Other	Academic Coaching / Advising, Vocational Support, Social Skills Interventions, Psychological / Mental Health Services, Communication with campus / program staff, Independent Living Support, Tutoring	Yes, 2 Year / Associate's Degree
https://www.kutztown.edu/about-ku/administrative-offices/disability-services-(dso)/my-place-program-(autism-services).html	ASD	Yes	High School Transcript, Interview	ASD Diagnosis, High school IEP, Psychological Evaluation at the University	$2500	8	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Social Events, Study Hall, Physical Fitness, Access to sensory reduced areas	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.landmark.edu/admissions/tac	ADHD	Yes	Unknown	Unknown	$3750	2	Unknown	Academic Coaching / Advising, Mentor Check-Ins, Social Events, Parental Support, Life Coaching, Special Courses, Pre-Semester Orientation / Transition Program	No
https://www.letu.edu/student-life/bridges-program.html	ASD	Yes	Writing Sample	ASD Diagnosis	$4000	N/A	Unknown	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Communication with campus / program staff, Study Hall, Pre-Semester Orientation / Transition Program, weekly support group,	Yes, 4 Year / Bachelor's Degree
https://www.lipscomb.edu/academics/undergraduate-studies/ideal-program/program-details	IDD	Yes	Unknown	Unknown	$8188	N/A	Unknown	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Physical Fitness, Independent living skills	No
https://www.loras.edu/academics/academic-support/lynch-learning-center/arch/	ASD	Yes	Unknown	ASD Diagnosis		N/A	Other	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Mentor Check-Ins, Social Events, Group Psychological Counselling, Tutoring, Special Courses, Pre-Semester Orientation / Transition Program, access to a sensory reduced space	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.marian.edu/campus-life/personalized-learning-center/autism-program	ASD, Neurodiversity Generally	Yes	Interview	Self-Identification	$6000	8	Other	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Social Events, Communication with campus / program staff, Study Hall, Life Coaching, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree
https://www.marquette.edu/disability-services/on-your-marq/	Neurodiversity Generally	Yes	Interview, Writing Sample	High school IEP, Psychological Evaluation at the University	$3000	8	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Social Skills Interventions, Psychological / Mental Health Services, Independent Living Support, Tutoring, weekly seminars	Yes, 4 Year / Bachelor's Degree
https://www.marshall.edu/collegeprogram/	ASD	Yes	High School Transcript, Writing Sample	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University, IQ test, Achievement Testing	$5000	N/A	Unknown	Academic Coaching / Advising, Vocational Support, Internship, Job Placement Services, Mentor Check-Ins, Social Skills Interventions, Social Events, Individual Psychological Counselling, Psychological / Mental Health Services, Communication with campus / program staff, Independent Living Support, Life Coaching, Relationship / Sexuality / Gender Identity Support, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.mercyhurst.edu/academics/autism-initiative-mercyhurst	ASD	Yes	Interview	Unknown		N/A	Other	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Job Placement Services, Mentor Check-Ins, Social Skills Interventions, Social Events, Independent Living Support, Study Hall, Life Coaching, Special Courses, Relationship / Sexuality / Gender Identity Support, Pre-Semester Orientation / Transition Program, Designated Dorm	Yes, 4 Year / Bachelor's Degree
https://www.millersville.edu/programs/integrated-studies.php	IDD	Yes	Writing Sample, Letter of Recommendation	High school IEP, Psychological Evaluation outside the University, Have a documented diagnosis of an intellectual and developmental disability as defined by the American Association on Intellectual and Developmental Disabilities		8	Other	Academic Coaching / Advising, Vocational Support, Internship, Special Courses, Not a lot of information about supports	No
https://www.misericordia.edu/campus-community/clinics-centers/autism/integrated-studies	ASD, ID	Yes	High School Transcript, Writing Sample, Letter of Recommendation	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University		4	Research Lab	Vocational Support, Internship, Social Skills Interventions, Special Courses, Not a lot of information about supports. Non-degree program where students take a course of study that involves inclusive classes	No

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.mmm.edu/offices/academic-access-program/	Learning Disabilities	Yes	Interview, Writing Sample	Psychological Evaluation at the University, comprehensive IQ test, Achievement testing, testing of language skills, executive function assessment. very stringent testing guidelines, outlined in detail on website	$3000	8	Other	Academic Coaching / Advising, Social Skills Interventions, Tutoring, Testing Accomodations, Priority registration	Yes, 4 Year / Bachelor's Degree
https://www.mtsac.edu/access/puzzle_project/puzzle_project.html	ASD	Yes	Interview	ASD Diagnosis		4	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Social Skills Interventions, Psychological / Mental Health Services, Tutoring, Relationship / Sexuality / Gender Identity Support, supports offered as a weekly series of workshops	Yes, 2 Year / Associate's Degree
https://www.muskingum.edu/plus	Neurodiversity Generally	Yes	Interview	High school IEP, Psychological Evaluation outside the University	$4750	N/A	Other	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Social Skills Interventions, Life Coaching, Tutoring, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree
https://www.mville.edu/life-manhattanville/student-services/center-student-accommodations/prospective-students-and-1	ASD	Yes	Unknown	Unknown	$5650	N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Skills Interventions, Social Events, Group Psychological Counselling, Tutoring	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.newhaven.edu/student-life/diversity-inclusion/accessibility-resources-center/more-than-access-faq.php	ASD	Yes	Letter of Recommendation	High school IEP, Psychological Evaluation outside the University	$0	N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Social Skills Interventions, Psychological / Mental Health Services, Communication with campus / program staff	Yes, 4 Year / Bachelor's Degree
https://www.newpaltz.edu/drc/asdservices.html	ASD	No	Unknown	Unknown		N/A	Disability Services Office	Academic Coaching / Advising, Vocational Support, Special Courses, not a lot of information online	Yes, 4 Year / Bachelor's Degree
https://www.nicholls.edu/education/support-programs/bridge-to-independence/	ASD, ID	Yes	High School Transcript, Interview	High school IEP, Psychological Evaluation outside the University		8	Unknown	Academic Coaching / Advising, Peer Mentoring, Internship, Social Skills Interventions, Social Events, Communication with campus / program staff, Tutoring	Yes, 4 Year / Bachelor's Degree
https://www.nova.edu/accessplus/index.html	ASD	Yes	High School Transcript, Interview, Writing Sample	ASD Diagnosis, Psychological Evaluation outside the University, IQ and Achievement tests (WISC, Stanford Binet, Woodcock-Johnson)	$9090	N/A	Unknown	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Social Events, Group Psychological Counselling, Independent Living Support, Study Hall, On Call Support, Designated Dorm	Yes, 4 Year / Bachelor's Degree
https://www.ohio.edu/accessibility/autism-spectrum-peer-coaching-team-aspect	ASD	Yes	Interview	ASD Diagnosis, Psychological Evaluation outside the University		N/A	Disability Services Office	Peer Mentoring	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.pace.edu/oasis	ASD, Neurodiversity Generally	Yes	High School Transcript, Writing Sample, Letter of Recommendation	High school IEP, Psychological Evaluation outside the University	$7000	N/A	Unknown	Academic Coaching / Advising, Vocational Support, Internship, Social Skills Interventions, Psychological / Mental Health Services, Communication with campus / program staff, Independent Living Support, Testing Accomodations	Yes, 4 Year / Bachelor's Degree
https://www.purchase.edu/offices/disability-resources/cornerstone-connect/	Not explicitly stated	Yes	Unknown	Unknown		N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Social Skills Interventions, Social Events, Life Coaching	Yes, 4 Year / Bachelor's Degree
https://www.ramapo.edu/enhance/	ASD	Yes	High School Transcript, Interview, Writing Sample	Psychological Evaluation outside the University	$3,587.50	N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Group Psychological Counselling, Communication with campus / program staff, Independent Living Support, Parental Support, Tutoring, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree
https://www.rit.edu/spectrumsupport/	ASD	Yes	Interview	None	$2250	N/A	Unknown	Academic Coaching / Advising, Vocational Support, Job Placement Services, Mentor Check-Ins, Social Skills Interventions, Social Events, Communication with campus / program staff, Life Coaching, Special Courses, On Call Support, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://sites.rowan.edu/path/	ASD	No	Required "registration form" not application	None	$0	N/A	Unknown	Academic Coaching / Advising, Vocational Support, Internship, Job Placement Services, seems like mostly a job placement program	Yes, 4 Year / Bachelor's Degree
https://rcaas.rutgers.edu/CSP	ASD	Yes	High School Transcript, Interview	ASD Diagnosis, High school IEP, Psychological Evaluation at the University	$3500	8	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Job Placement Services, Mentor Check-Ins, Social Skills Interventions, Social Events, Psychological / Mental Health Services, Communication with campus / program staff, Study Hall, Life Coaching	Yes, 4 Year / Bachelor's Degree
https://mainstay-sails.seattlecentral.edu/sails-college-students/steps-enroll	ASD, Neurodiversity Generally	Yes	Interview	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University, any of the above	$3450	N/A	Unknown	Academic Coaching / Advising, Vocational Support, Job Placement Services, Communication with campus / program staff, Tutoring, Testing Accomodations	Yes, 2 Year / Associate's Degree
https://seu.edu/academics/ese/link/	IDD	Yes	Unknown	Unknown	$8500	4	Unknown	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Social Events, Independent Living Support, Special Courses, Designated Dorm	No
https://www.sju.edu/kinney/autism-services/college-student-support-services	ASD	Yes	Unknown	Unknown	$4000	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Social Events, Communication with campus / program staff, Parental Support, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.snc.edu/counseling/autism.html	ASD	No	No Application Required	ASD Diagnosis	$0	8	Counselling or Psychological services office	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Psychological / Mental Health Services, Communication with campus / program staff	Yes, 4 Year / Bachelor's Degree
https://www.sru.edu/academics/academic-services/services-for-students-with-disabilities/rock-life-program	ASD, ID	Yes	High School Transcript, Interview, Writing Sample, Letter of Recommendation	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University	$11,392	8	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Social Events, Independent Living Support, Special Courses, Physical Fitness, Designated Dorm	No
https://med.stanford.edu/neurodiversity/NSSP-temp/neurodiverse-student-support-program/peer-mentor-program.html	Neurodiversity Generally	Yes	Unknown	None	$0	2	Research Lab	Peer Mentoring, Vocational Support, Social Events, Psychological / Mental Health Services, Communication with campus / program staff, Independent Living Support, Special Courses, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree
https://www.stcloudstate.edu/sassi/default.aspx	ASD, Neurodiversity Generally	Unknown	Unknown	Unknown	$0	N/A	Unknown	Peer Mentoring, Speech / Language Interventions, group support meetings	Yes, 4 Year / Bachelor's Degree
https://www.taftcollege.edu/til/	IDD	Yes	High School Transcript, Interview, Letter of Recommendation	High school IEP, Psychological Evaluation outside the University		4	Unknown	Vocational Support, Internship, Independent Living Support, Life Coaching, Special Courses, Designated Dorm	No

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://aggieachieve.tamu.edu/	IDD	Yes	High School Transcript, Letter of Recommendation, Application Fee, video essay	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University, ID Diagnosis, psych eval including an achievement test	$8563	8	Academic Department	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Independent Living Support, Tutoring, Special Courses, Pre-Semester Orientation / Transition Program	No
https://www.towson.edu/iwb/centers/hussman/programs/documents/caps-information.pdf	ASD	No	No Application Required	Unknown	$1050	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Social Skills Interventions, Life Coaching, group peer mentor meetings	Yes, 4 Year / Bachelor's Degree
https://www.depts.ttu.edu/burkhartcenter/case/	ASD, ID	Yes	High School Transcript, Interview, Writing Sample, Letter of Recommendation	Psychological Evaluation outside the University		N/A	Research Lab	Academic Coaching / Advising, Vocational Support, Internship, Mentor Check-Ins, Social Skills Interventions, Psychological / Mental Health Services, Communication with campus / program staff, Independent Living Support, Life Coaching, Tutoring	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://ua-acts.as.ua.edu/	ASD	Yes	High School Transcript, Interview, Letter of Recommendation	ASD Diagnosis, Psychological Evaluation outside the University, IQ and academic skills in psych eval	$3600	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Skills Interventions, Social Events, Individual Psychological Counselling, Psychological / Mental Health Services, Communication with campus / program staff, Independent Living Support, Study Hall, Parental Support, On Call Support	Yes, 4 Year / Bachelor's Degree
https://empower.uark.edu/	Cognitive Disabilities	Yes	High School Transcript, Interview, Writing Sample, Letter of Recommendation	High school IEP, Psychological Evaluation outside the University, documentation of a cognitive disability	$5000	8	Academic Department	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Mentor Check-Ins, Individual Psychological Counselling, Psychological / Mental Health Services, Independent Living Support, Special Courses	No
https://uca.edu/dst/autism-advocacy-program/	ASD, Neurodiversity Generally	Yes	Interview, Writing Sample	High school IEP		N/A	Other	Academic Coaching / Advising, Mentor Check-Ins, Social Events, Individual Psychological Counselling, Communication with campus / program staff, Tutoring, Special Courses, Low-sensory registration day for entering freshmen and early registration for returning students, occupational therapy, low-sensory area	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://redwoodseed.ucdavis.edu/	IDD	Yes	High School Transcript, Interview, Letter of Recommendation, Application Fee, skill survey	High school IEP, Psychological Evaluation outside the University	$8,799.50	8	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Social Skills Interventions, Psychological / Mental Health Services, Communication with campus / program staff, Independent Living Support, Life Coaching, Tutoring, Special Courses, Physical Fitness, 1:1 Help in Class	No
https://www.cds.udel.edu/spectrum-scholars/	ASD	Yes	Unknown	ASD Diagnosis		N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Job Placement Services, Mentor Check-Ins, Independent Living Support, Life Coaching, Special Courses, Pre-Semester Orientation / Transition Program, group workshops, partnered with JP Morgan Chase	Yes, 4 Year / Bachelor's Degree
https://www.uhcl.edu/autism-center/connecting-to-college/	ASD	Unknown	Unknown	Unknown	$0	N/A	Research Lab	Academic Coaching / Advising, Vocational Support, Mentor Check-Ins, Social Events, Psychological / Mental Health Services, Communication with campus / program staff, Independent Living Support	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.uidaho.edu/current-students/cdar/raven-scholars	ASD	Yes	Unknown	None	$0	8	Unknown	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Skills Interventions, Social Events, Communication with campus / program staff, Life Coaching, Special Courses, workshops	Yes, 4 Year / Bachelor's Degree
https://signa.umd.edu/	Neuro-diversity Generally	Yes	Unknown	Unknown	$2000	4	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Skills Interventions, Social Events, Relationship / Sexuality / Gender Identity Support	Yes, 4 Year / Bachelor's Degree
https://www.uml.edu/research/autism/services/horizons.aspx	ASD	Yes	Unknown	Unknown	$75	8	Research Lab	Peer Mentoring, Social Skills Interventions, Group Meetings with students and peer mentors	Yes, 4 Year / Bachelor's Degree
https://www.umsl.edu/succeed/	IDD	Yes	High School Transcript, Letter of Recommendation, Application Fee	ASD Diagnosis, High school IEP	$5250	4	Unknown	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Job Placement Services, Mentor Check-Ins, Study Hall, Parental Support, Tutoring, Special Courses, Testing Accomodations	Yes, 4 Year / Bachelor's Degree
https://www.umt.edu/speech-language-hearing-occupational-sciences/dewit-ritecare-clinic/mossaic.php	ASD	Yes	Unknown	Unknown		N/A	Research Lab	Peer Mentoring, Social Skills Interventions, Social Events, Speech / Language Interventions, Weekly Group Meeting with mentors and students	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

ASP Website	Population served	Program application	Elements of application	Diagnostic Criteria of application	Cost per semester	Duration of support (Number of semesters)	Administrative home	Program supports offered	Degree eligibility
https://www.utc.edu/enrollment-management-and-student-affairs/disability-resource-center/mosaic	ASD	Yes	High School Transcript, Interview, Letter of Recommendation	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University	$3500	8	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Mentor Check-Ins, Social Skills Interventions, Study Hall, Special Courses, Relationship / Sexuality / Gender Identity Support	Yes, 4 Year / Bachelor's Degree
https://newstudentservices.utexas.edu/longhorn-ties/student-support#comprehensive	Neurodiversity Generally	Yes	Unknown	Self-Identification		N/A	Disability Services Office	Academic Coaching / Advising, Mentor Check-Ins, Social Skills Interventions, Social Events, Study Hall, Life Coaching, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree
https://futureut.utk.edu/	IDD	Yes	High School Transcript, Interview, Letter of Recommendation	High school IEP, Psychological Evaluation outside the University, Achievement Test	$10,000	6	Academic Department	Peer Mentoring, Vocational Support, Internship, Independent Living Support, Life Coaching, Special Courses	No
https://accessibility.utulsa.edu/students/pathfinders/	ASD	No	No Application Required	None		N/A	Disability Services Office	Academic Coaching / Advising, Social Events, Relationship / Sexuality / Gender Identity Support, skills-building groups	Yes, 4 Year / Bachelor's Degree
https://www.uvu.edu/autism/passages/	ASD	Yes	Writing Sample	ASD Diagnosis, High school IEP	$600	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Special Courses	Yes, 4 Year / Bachelor's Degree
https://sphsc.washington.edu/mossaic	ASD	Unknown	Unknown	None		N/A	Academic Department	Peer Mentoring, Mentor Check-Ins, Social Events, Communication with campus / program staff	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

https://www.uvu.edu/autism/passages/	ASD	Yes	Writing Sample	ASD Diagnosis, High school IEP	$600	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Special Courses	Yes, 4 Year / Bachelor's Degree
https://uwf.edu/centers/center-for-behavior-analysis/argos-autism-program/	ASD	Unknown	Unknown	Unknown		N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Social Events, Communication with campus / program staff, Tutoring, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree
https://peabody.vanderbilt.edu/departments/nextsteps/	IDD	Yes	Unknown	Psychological Evaluation outside the University	$12,354	8	Research Lab	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Internship, Mentor Check-Ins, Social Events, Tutoring, Special Courses, Physical Fitness, Pre-Semester Orientation / Transition Program, Designated Dorm	No
https://www.wcmo.edu/academics/ssc/loc/college-transition/index.html	ASD	Yes	High School Transcript, Interview, Letter of Recommendation	Psychological Evaluation outside the University, achievement tests	$2000	8	Disability Services Office	Academic Coaching / Advising, Vocational Support, Internship, Mentor Check-Ins, Social Skills Interventions, Psychological / Mental Health Services, Communication with campus / program staff, Study Hall, Parental Support, Physical Fitness, Testing Accomodations, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree
https://www.wcupa.edu/dcap/	ASD		Standardized Test Scores, Writing Sample	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University	$2500	8	Other	Academic Coaching / Advising, Vocational Support, Social Skills Interventions, Social Events, Independent Living Support, Study Hall, Life Coaching, group workshops	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

https://www.uvu.edu/autism/passages/	ASD	Yes	Writing Sample	ASD Diagnosis, High school IEP	$600	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Special Courses	Yes, 4 Year / Bachelor's Degree
https://www.wku.edu/kapcircleofsupport/	ASD	Yes	High School Transcript, Standardized Test Scores, Interview, Letter of Recommendation	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University	$5000	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Individual Psychological Counselling, Group Psychological Counselling, Psychological / Mental Health Services, Study Hall, Tutoring, guarantee of single dorm room	Yes, 4 Year / Bachelor's Degree
https://www.wm.edu/sites/neurodiversity/wmbridge/index.php	Neurodiversity Generally	No	No Application Required	None	$0	N/A	Research Lab	Social Events, Pre-Semester Orientation / Transition Program, Neurodiversity Student Group, Workshops/ Presentations	Yes, 4 Year / Bachelor's Degree
https://wmich.edu/autism/asc	ASD	Yes	High School Transcript, Interview, Writing Sample	None	$2000	N/A	Research Lab	Academic Coaching / Advising, Vocational Support, Job Placement Services, Mentor Check-Ins, Social Events, Communication with campus / program staff, Parental Support, Pre-Semester Orientation / Transition Program, weekly workshops	Yes, 4 Year / Bachelor's Degree
https://www.goldenwestcollege.edu/dsps/asd/index.html	ASD	No	No Application Required	ASD Diagnosis	$0	4	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Social Events, Parental Support, Testing Accomodations, Relationship / Sexuality / Gender Identity Support, weekly workshops	Yes, 2 Year / Associate's Degree
https://www.wright.edu/disability-services/autism-spectrum-support	ASD	Yes	Interview	ASD Diagnosis	$750	2	Disability Services Office	Peer Mentoring, Mentor Check-Ins	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

https://www.uvu.edu/autism/passages/	ASD	Yes	Writing Sample	ASD Diagnosis, High school IEP	$600	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Special Courses	Yes, 4 Year / Bachelor's Degree
https://www.xavier.edu/accessibility-and-disability-resources/x-path-program/index	ASD	Yes	Interview, Writing Sample	ASD Diagnosis, High school IEP, Asperger's or PDD-NOS also accepted	$1,500	N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Events, Study Hall, Life Coaching, Peer Support Group, work shops	Yes, 4 Year / Bachelor's Degree
https://connections.calpoly.edu/	ASD	No	No Application Required	None	$0	N/A	Disability Services Office	Weekly Peer Support Group	Yes, 4 Year / Bachelor's Degree
https://www.cbc.edu/disabilitysupport	ASD	Yes	Standardized Test Scores, Letter of Recommendation	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University	$500	2	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Group Psychological Counselling	Yes, 4 Year / Bachelor's Degree
https://www.cuny.edu/current-students/student-affairs/student-services/disability/#programs	ASD	Unknown	Unknown	Unknown		N/A	Disability Services Office	Peer Mentoring, Mentor Check-Ins, group mentor meetings	Yes, 4 Year / Bachelor's Degree
https://med.emory.edu/departments/psychiatry/autism/services/oaks/index.html	ASD	Yes	Unknown	ASD Diagnosis	$3500	N/A	Research Lab	Academic Coaching / Advising, Mentor Check-Ins, Individual Psychological Counselling, Psychological / Mental Health Services, Communication with campus / program staff	Yes, 4 Year / Bachelor's Degree
http://fairmontstate.edu/schoolofeducation/aimss	ASD	Yes	Interview, Writing Sample	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University, IQ and Achievement Testing	$5000	N/A	Academic Department	Academic Coaching / Advising, Peer Mentoring, Social Skills Interventions, Social Events, Independent Living Support, Study Hall	Yes, 4 Year / Bachelor's Degree
https://www.merrimack.edu/academics/academic-success-center/accessibility-services/the-activate-program/	ASD, Neurodiversity Generally	Unknown	Unknown	Self-Identification		N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Special Courses	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

https://www.uvu.edu/autism/passages/	ASD	Yes	Writing Sample	ASD Diagnosis, High school IEP	$600	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Special Courses	Yes, 4 Year / Bachelor's Degree
https://www.rcpd.msu.edu/programs/signature-programs/bond	ASD	Unknown	Unknown	Unknown		N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Mentor Check-Ins, Social Skills Interventions, Social Events, Special Courses, Pre-Semester Orientation / Transition Program	Yes, 4 Year / Bachelor's Degree
https://www.addc.msstate.edu/liaison	ASD	Unknown	Unknown	Unknown		N/A	Research Lab	Academic Coaching / Advising, Vocational Support, Mentor Check-Ins, Social Events, Psychological / Mental Health Services, Communication with campus / program staff, Tutoring	Yes, 4 Year / Bachelor's Degree
https://www.nyu.edu/students/communities-and-groups/student-accessibility/nyu-connections-asd-program.html	ASD	Unknown	Unknown	Unknown	$0	N/A	Disability Services Office	Academic Coaching / Advising, Mentor Check-Ins, Social Events, Peer Group Meetings	Yes, 4 Year / Bachelor's Degree
https://nisonger.osu.edu/adolescent/ace-autism-college-experience-at-osu/	ASD	Unknown	Unknown	Self-Identification	$4000	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Social Skills Interventions, Social Events, Communication with campus / program staff, Study Hall	Yes, 4 Year / Bachelor's Degree
https://www.reinhardt.edu/wp-content/uploads/2020/06/SEAD.pdf	ASD, Learning Disabilities	Unknown	Unknown	Unknown		N/A	Disability Services Office	Academic Coaching / Advising, Social Skills Interventions, Social Events	Yes, 4 Year / Bachelor's Degree
https://www.stmarys-ca.edu/student-disability-services	ASD, Neurodiversity Generally	No	No Application Required	Unknown	$0	N/A	Disability Services Office	Academic Coaching / Advising, Peer Mentoring, Vocational Support, Social Events, Workshops	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

https://www.uvu.edu/autism/passages/	ASD	Yes	Writing Sample	ASD Diagnosis, High school IEP	$600	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Special Courses	Yes, 4 Year / Bachelor's Degree
https://www.sfcollege.edu/drc/sos/index	ASD	Yes	Writing Sample	ASD Diagnosis		N/A	Disability Services Office	Social Events, Communication with campus / program staff, Parental Support, Peer Group Meetings	Yes, 4 Year / Bachelor's Degree
https://www.seminolestate.edu/dss/disability-type/autism-spectrum-disorders	ASD	Unknown	Unknown	Unknown		8	Disability Services Office	Social Skills Interventions, Independent Living Support	Yes, 4 Year / Bachelor's Degree
https://www.tccd.edu/academics/ce/preparatory-education/transitional-skills/autism-spectrum-disorder-program/	ASD, ID	Yes	Unknown	Unknown			Other	Academic Coaching / Advising, Vocational Support, Social Skills Interventions, Communication with campus / program staff, Independent Living Support, Special Courses, Peer Group Meetings	No
https://www.tiu.edu/undergrad/academic-support-resources/	ASD	Unknown	Unknown	Unknown		N/A	Unknown	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Skills Interventions	Yes, 4 Year / Bachelor's Degree
https://disability.ufl.edu/get-involved/social-gators/	ASD	Yes	Writing Sample	Self-Identification	$0	N/A	Disability Services Office	Peer Mentoring, Mentor Check-Ins, Weekly Peer Group Meeting / Workshop	Yes, 4 Year / Bachelor's Degree
https://www.unf.edu/thrive/	ASD	Yes	Interview	ASD Diagnosis	$0	8	Disability Services Office	Academic Coaching / Advising, Vocational Support, Mentor Check-Ins, Social Events, Communication with campus / program staff, Parental Support, Tutoring	Yes, 4 Year / Bachelor's Degree
https://neurodiversity.unt.edu/unt-engage	Neurodiversity Generally	Yes	High School Transcript, Interview, Writing Sample, Letter of Recommendation	High school IEP, Psychological Evaluation outside the University		8	Other	Academic Coaching / Advising, Mentor Check-Ins, Group Psychological Counselling, Life Coaching, Tutoring	Yes, 4 Year / Bachelor's Degree

Table 3 (continued)

https://www.uvu.edu/autism/passages/	ASD	Yes	Writing Sample	ASD Diagnosis, High school IEP	$600	N/A	Research Lab	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins, Social Events, Special Courses	Yes, 4 Year / Bachelor's Degree
https://web.uri.edu/disability/s-t-a-r-t-uri/	ASD	Yes	Interview	Unknown	$0	2	Disability Services Office	Vocational Support, Internship, Social Skills Interventions, Social Events, Special Courses, Pre-Semester Orientation / Transition Program, Workshops	Yes, 4 Year / Bachelor's Degree
https://usao.edu/future-students/neill-wint-center-for-neurodiversity.html	ASD	Yes	Standardized Test Scores, Writing Sample, Letter of Recommendation	ASD Diagnosis, High school IEP, Psychological Evaluation outside the University	$1000	N/A	Other	Academic Coaching / Advising, Peer Mentoring, Mentor Check-Ins	Yes, 4 Year / Bachelor's Degree
https://uaptc.edu/3D	IDD	Yes	High School Transcript, Interview, Writing Sample, Letter of Recommendation	High school IEP, Psychological Evaluation outside the University, IQ scores and Achievement tests	$2254	6	Academic Department	Special Courses, Technical Certificate program in Baking, Culinary or Hospitality specifically for people with IDD	No

Data gathered in April 2023

Application Process

While the number of offerings is expanding, major barriers to entry remain for students with ASD. First, all identified programs which serve degree seeking students require that the student first be admitted to the institution before applying for an ASP. This may be a barrier to access for students with ASD who are more likely to struggle with reading, writing, and mathematical problem solving (Fleury et al., 2014) and may not meet academic admissions requirements. Similarly, students with ASD are more likely to perform poorly on standardized testing required for admission (VanBergeijk et al., 2008). Further, this admission process suggests that institutions with ASPs are not actively collaborating with secondary education providers (e.g., high schools) to identify students with autism interested in attending the institution who may also benefit from ASP services. An alternative process would enable simultaneous admission to both the institution and the ASP. This would ensure that students who need ASP services get them from the outset of their education, in contrast to the reactive, "wait-to-fail" model of support which may be commonplace at some institutions (A. Anderson, 2018). Additionally, a simultaneous admissions process could help institutions identify and provide support to promising students who might not meet standard admissions criteria.

For students who gain admission to an institution, we found that a large majority (74.4%) of programs require a further program specific application. This application is highly likely (98%) to require an interview with the student. While it is reasonable to assume that program staff are aware of the social communication difficulties of individuals with ASD, many students are likely to experience heightened anxiety during an interview (Spain et al., 2018) and thus may need to prepare for this part of the application process. Other common aspects of program specific applications are high school transcripts (94%), a writing sample (64%), letters of recommendation (58%) and standardized test scores (15%). In addition to the program application, most programs require some proof of diagnosis, with some programs requiring multiple forms of documentation. This may be a significant barrier to access as a large portion of individuals with ASD lack a formal diagnosis (Kim et al., 2011; Pilling et al., 2012) and therefore cannot document they have an ASD. Prospective students and their families can also expect to need a high school IEP and a recent (within 3 years) psychological evaluation as part of their application packet. Some programs further require IQ and achievement testing as a part of the psychological evaluation. Given that many secondary students with ASD experience inadequate post-secondary transition planning (Brunton, 2018) individuals and families may be unable to provide this documentation.

Cost

Cost is another potential barrier to entry for prospective students, as most programs for which data was available (78%) operate on a fee for service model. These fees are charged in addition to undergraduate tuition, exacerbating the cost of attending college. Similar to previous works, we found that the cost of participation varies greatly between programs within $75 to $19,640 per semester (Barnhill, 2016; Viezel, et. al., 2020). Barnhill reports an average cost of $3,262.5 per semester and Viezel reports an average cost of $3,338.44 per semester. We found the average cost of participation to be $4,603 per semester with a median of $3,670 per semester, representing a 37.9% increase in average cost between our work and the costs reported by Viezel. Partially, this increase in cost can be explained by the effect of inflation since the previous data was recorded (approximately 24%) and some variability from our evaluation of a larger number of programs. However, the varied administrative structure of the programs themselves is also a factor. For example, the most expensive program, at a cost of $19,640 per semester, is a non-degree granting program in which students pay only for participation in the program, not undergraduate tuition.

Generally, we find that ASP cost is expensive in both absolute terms and relative to median income. Figure 5 shows the comparison and correlation between the cost of ASPs and the median income of their states, separated by public and private institutions. Additionally, the cost of participation in ASPs is negatively correlated with the median income of the state in which the institution resides. Particularly troubling is that ASP services can be expensive at both private and public institutions, and there appears to be little difference in overall cost of ASPs at public universities, which disproportionately serve students from lower- to middle- income families (Chen, 2019). Consequently, students from lower- to middle- income families may disproportionately bear the burden of ASP fees. Given ever increasing tuition at U.S. colleges and universities (Mitchell et al., 2019), ASP fees are likely to increase financial burdens on students and their families or render participation in these programs impossible due to their cost. Future work should evaluate the availability of financial aid and scholarships which could lessen the financial burden on these students and their families.

Administrative Home of the Program

The administrative home of ASPs highlights the high level of variability across institutions; in keeping with the great diversity of organizational structures found across colleges and universities. This variability suggests that there is not

a standard model or "best way" for the manner in which these programs function within postsecondary institutions. This may be taken as a positive indicator that these institutions are organizing their ASPs administratively in a manner that best aligns the ASP with the institution's overall goals. But this may also be problematic because ASPs may be "hidden" within administrative units of the university not typically associated with disabilities services, and therefore challenging for students to locate.

Notably, our results contrast with previous research which reported counseling or psychological services offices as the most common administrative housing (Viezel, et. al., 2020). It is unclear if this difference is due to the difference in sampling of programs or if some programs have migrated to other administrative housing in the time between the two works. Additionally, administrative variability may indicate that ASPs are targeting certain constituencies of students over others. As Fig. 6 shows, nearly 70% of ASPs were located outside of the school's disability services office, or in an unknown location. While a standard of administrative housing and function may never emerge the distribution of administrative housing sheds light on which areas of the university have taken an interest in supporting students with ASD, rather than creating a pro forma office that may be mis-aligned with the rest of the institution.

Institution Type and Control

We find that ASPs occur across different forms of institution type and control, but their occurrence is not proportional to the total number of these institutions. Previous research found that between 63 and 71% of ASPs occur at public institutions (Barnhill, 2016; Viezel, et. al., 2020; Nachman, et, al., 2021), Similarly, we found a majority (59%) of ASPs occur at public, 4-year institutions, despite the fact that these types of institutions account for only 21% of all colleges and universities in the U.S. Nearly all (94%) of the ASPs we identify are hosted at 4-year institutions, representing a major gap in service at 2-year institutions. Of the few programs which are hosted at 2-year institutions, only half of them offer students the opportunity to earn credits towards a degree. This presents an accessibility problem for autistic students who wish to seek a degree but may not be able to meet the admission requirements of a 4-year institution which offers an ASP. If these students wish to pursue a degree at a 2-year institution, they will likely struggle to find one which offers an ASP. Notably, of the 16 ASPs which we found to no longer be active 14 of them had been hosted at public, 4-year institutions. While we cannot say why these programs are no longer active, it is possible that they relied on and lost access to some form of public funding.

Support Offered

The nature of support offered by ASPs also varies widely between institutions, but there are some common trends. As identified in previous works (Viezel et al., 2020), academic coaching, peer mentoring, vocational support, social events, and mentor check-ins are the most common supports offered to program participants. These supports suggest that the aim of these programs is to connect students with individuals whom they can meet with regularly to help them through their transition to college life. Program staff, who serve as academic coaches and mentors, assist the student in navigating the academic and bureaucratic aspects of college life. Peer mentors and program sponsored social events help students to socially integrate into the campus community. ASPs support the health and well-being of students through mental health services, individual and/or group counselling, and physical fitness programs. This combination of social and psychological support addresses one of the primary concerns that literature has identified among autistic college students (Anderson et al., 2017; Davis et al., 2021). The programs also aim to support students transition into the workforce post-graduation through vocational supports, such as interview preparation and resume writing, access to internships on and off campus, and job placement services. This is especially important for this population, as research has shown that adults with ASD have lower rates of employment than adults without disabilities or adults with other disabilities (Burgess & Cimera, 2014; Nord et al., 2016; Roux et al., 2013).

Degree Offerings

Finally, we found that most of the programs identified (79.7%) support degree seeking students, while the minority of programs offer certificates of completion to non-degree students. These programs are distinct in the application process; degree seeking programs require admission to the university and often require a program application, while non-degree programs do not require admission to the university but always require a program application. Additionally, many non-degree programs are designed to serve students with IDD, who may have co-occurring ASD, while degree-seeking programs are designed for students with ASD who do not have co-occurring IDD.

Discoverability and Missing Data

The basis of the present work is the use of data science techniques to discover ASPs in the U.S. The necessity of these techniques to achieve a comprehensive accounting of available programs is troubling for stakeholders in this space. A prospective student cannot be expected to go to the lengths that the researchers did to obtain a robust understanding of the supports available to them in postsecondary education. The first issue that a prospective student will encounter is the difficulty in determining which of the thousands of postsecondary institutions hosts an ASP. As shown in Fig. 2, even with a specially crafted search-engine query a student would have to evaluate up to 50 results to discover some ASPs. Research has shown that click through rate for search engine results drops off significantly after the first result (Craswell et al., 2008). Once students have discovered the webpage for a specific ASP, they may still struggle to determine if that program will meet their needs. As shown in Table 2, the support offered varies widely by program. Additionally, as shown in Table 3, missing data is a persistent issue. In cases where information is available, it may require a large effort to discover.

ASPs exist to benefit students, so it seems that prospective students and their parents are responsible for locating an ASP. However, ASPs are responsible for making their existence discoverable. Information Foraging Theory states that foragers look for cues, and in the case of Internet searches, "information scents," cues derived from links, labels, contexts, and experience (Pirolli, 2007), when deciding to pursue further information on a webpage. Similarly, frugivores scan for visual and olfactory cues while foraging. Evolutionary biologists found that fruits and flowers have evolved color contrasts and scents that attract these foragers – in a sense the fruits "want" to be eaten so that their seeds will be distributed (Nevo et al., 2018a, 2018b) just as flowers "want" to be visited by pollinators so that their pollen will spread (Rudall, 2020). If ASPs "want" to be found, they may have to adopt similar strategies as fruits and flowers – that is cues that will make them stand out in an internet search. Search engine optimization (SEO) practices (Sharma et al., 2019) increase the quantity and quality of organic (as opposed to paid placement) search results through such practices as keywords, metatags, structured data, and HTML tags. We suggest that an information foraging perspective places equal responsibility on the prospective student and the ASP for discoverability. Although our data science based approach uncovered many more ASPs than manual internet search reveals and our results are made available with this paper, the changing landscape of higher education and ASPs necessitates repeated searching. While it is not reasonable to expect that prospective students and parents will have the data science skills needed to replicate our search, it is reasonable for ASPs to evolve like fruits and flowers have and make their presence easily detectable through SEO practices.

Third party organizations that collect and host online resources which list ASP information, such as College Autism Network and College Autism Spectrum, can help to reduce the problem of discoverability (College Autism Network, 2025). Given the variability of support and service offerings among programs that we discovered, online resources should aim for comprehensive accounting of the characteristics of available programs. Additionally, these resources should include functionality to filter the list of programs based on the listed characteristics to accommodate a student population with varying needs. We would recommend that any organization hosting such a resource update the available information at least every school year, as new programs emerge, and existing programs may evolve their service offerings or cease to exist. To our knowledge, the aforementioned resources are regularly updated, however we presume these updates are done with manual procedures similar to those employed by previous research. Towards this end, these third party organizations may be the primary beneficiaries of our automated procedure. We hope that the approach for search and discovery we outlined here can make regular maintenance feasible while maintaining or improving the quantity of programs and quality of data reported about those programs. Furthermore, the researchers are aware of various paid college counseling services available for prospective autistic students. We hope that the reduction in labor required which results from our automation can make this vital information available to students free of charge. Future work may investigate the application of Information Extraction (Grisham, 2015; Small & Medsker, 2014) techniques to automate the process of determining the characteristics of individual ASPs.

Limitations

While our improved search technique has yielded positive results, there are potential limitations to our approach. First, we cannot conclusively say whether the increased number of ASPs that we identified compared to previous works is due to the superiority of our technique or a result of the emergence of new ASPs in the time between our and previous works. Our search was conducted in April 2023, while previous work was done in November 2018 (Viezel et al., 2020) and August 2019 (Nachman et al., 2021). Future works may repeat our data collection procedure and employ longitudinal data analysis to track the emergence of new ASPs and evolution of previously identified ASPs.

Second, our search technique leveraged the expertise of the researchers. It is not a viable option for prospective students who are seeking an ASP and does not address the problem of the difficulty of accessing information about ASPs that this population faces. Advancements in Generative Large Language Models (LLM) such as ChatGPT have made cutting edge AI technology more accessible for the general population. However, there are some limitations for use of these technologies in this space. For instance, LLMs are generally released with a knowledge cutoff which is up to two years old. Given the evolving nature of ASPs, LLMs may offer information which is no longer accurate. Future work may investigate the use of LLMs to assist prospective students but must contend with this issue of outdated training data inherent in the underlying model. Further, web content's dynamic and variable nature limits the effectiveness of any machine-learning-based approach. While our procedure discovered a large number of ASPs, it is highly likely that some were missed.

Finally, while we aimed to report as much information about the characteristics of ASPs as possible, we were limited by the persistent issue of missing data. The program webpages identified varied in the amount of information reported about the ASP, making it difficult for researchers and prospective students to evaluate the program. While our current methodology only analyzes web-based data, incorporating additional data sources may fill in some of the missing data. We are currently conducting an interview study with ASP staff, students, parents and high-school professionals to broaden our understanding of ASPs and the ways in which stakeholders interact with them. Future work should seek to further automate our search and classification pipeline and aim to develop a consumer-friendly format to disseminate the resulting information.

Conclusion

The present work contains, to our knowledge, the most comprehensive list of ASPs in the literature to date, including an accounting of the requirements and features of these programs. We found that the nature of ASPs across the U.S. varies widely between institutions. Nonetheless, our data highlights several consistent findings that underscore significant barriers for autistic students who seek access to higher education. First, all degree-granting ASPs require that students first be admitted to the institution before they can access the ASP's services. Second, many programs require a formal ASD diagnosis or other standardized evaluation, an application that can include an interview, writing sample, and letters of recommendation, and most programs charge a fee for participation. Since many autistic students may lack a formal diagnosis (Kim et al., 2011; Pilling et al., 2012), may not wish to disclose their autism to others (Anderson et al., 2018a, 2018b), or may lack the financial resources to pay for services, these are likely to be substantial barriers. Alternatively, institutions could explore ways to systematically identify autistic students who may benefit from supports but lack a formal diagnosis or other documentation, and institutions and policy makers could identify funding sources to subsidize student participation in ASPs, focusing on students with less access to financial resources.

Third, Table 2 and Fig. 7 show that many ASPs offer an array of academic and non-academic supports that align to the accommodation preferences of autistic college students (Accardo et al., 2019), and are offered over multiple semesters. However, our findings do not indicate how well the services offered align to the needs of individual students, including whether they support students with secondary diagnoses, or whether they lead to successful outcomes, measured by indicators such as GPA, 2-year and 4-year graduation rates, or measures of career attainment and psychosocial functioning. Clearly, more research is needed to establish which elements of ASP services are needed by the heterogeneous population of autistic college students, and which lead to the best outcomes.

As ASPs, and inclusion of students with ASD, are a dynamic issue in higher education, the information reported in this paper will be different soon. Further works may repeat our search process at a later date to discover newly emerging programs and may attempt to further automate the data extraction process to increase the feasibility of an annually updated catalog of ASPs, as well as serving as a growing historical document that reveals how ASPs and the institutions that house them continue to evolve. Indeed, this paper is not intended to be definitive but rather heuristic. We envision the present work as both a baseline for future researchers and policy makers, and a beta version of a future system that can be used to regularly update the data and offer it publicly for interested students, their families and researchers.

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Declarations

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